

SURVEY

Examining AI-Powered ADHD Diagnosis: Current Trends, Key Challenges, and Future Directions in the Field

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ABSTRACT Attention deficit hyperactivity disorder (ADHD) is a type of neurodevelopmental disease affecting the mental health of children and adults. Individuals with ADHD show various symptoms such as inattention, hyperactivity, and impulsivity. Early diagnosis of ADHD helps to alter neural connections and refine symptoms. The clinical practice to diagnose ADHD is through subjective measures and does not significantly capture the underlying structural and functional mechanisms of the brain. Therefore, it is crucial to explore other approaches such as Artificial Intelligence (AI) to improve the accuracy and efficacy of ADHD diagnosis. Consequently, in this article we systematically investigate various Machine Learning (ML) and Deep Learning (DL) approaches as well as different diagnostic tools or modalities employed for the identification of ADHD. Particularly, a Systematic Literature Review (SLR) is conducted to review and analyze 98 selected studies published from 2021 to 2024. Subsequently, the selected studies are grouped into five categories based on the modalities utilized in these studies: physiological signals (37), magnetic resonance imaging (31), questionnaires (11), motion data (8), and others (11). We also analyze AI models which indicates that 45 studies utilized ML models, 33 studies employed DL models, and 20 studies used both. However, there are still some gaps in current research such as a lack of publicly available datasets except MRI and EEG. Although datasets for MEG and actigraphy exist, but they are underexplored and have been utilized in only a few studies. While DL models like CNNs and ANNs have been increasingly applied in recent years for ADHD diagnosis, there is a shortage of advanced DL models, including transfer learning approaches like ResNet and VGG. Additionally, there is a lack of interpretability in AI models, particularly DL models. Furthermore, most studies focus on individual modalities for ADHD diagnosis, and despite many studies showing excellent results, there is a lack of implementation of AI-based tools in real-world clinical settings. These gaps highlight areas for further exploration and development.

INDEX TERMS ADHD, SLR, AI, ML, DL, physiological signals, MRI, motion data, questionnaires, EEG.

I. INTRODUCTION

Attention Deficit Hyperactivity Disorder (ADHD) is one of the most commonly diagnosed mental disorders. It is mainly developed in childhood, becomes more severe in adolescents, and persists into adulthood [1]. Almost 5-8 % of children/adolescents and 2.5 % of the adult population

affected by ADHD [2]. Studies reveal that in comparison to the general population, people suffering from ADHD have poor academic records, poor mental and physical health, are highly likely to face the risks of unemployment [3], generally experience low quality of life [4] and in severe condition can lead to the suicidal attempts [5].

ADHD patients mostly show three kinds of symptoms: inattention, impulsivity, and hyperactivity. However, this is a heterogeneous disorder, so individuals may

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display more or less of these individual symptoms. Therefore, ADHD is classified into three subtypes inattention, hyperactivity/impulsivity, and combined ADHD [6]. First, ADHD Inattentive (ADHD-I) is characterized by having inattentiveness and distractibility symptoms without hyperactivity/impulsivity symptoms. Second, ADHD hyperactivity/impulsivity (ADHD-H) is distinguished by having hyperactivity, impulsivity, and restlessness symptoms in the absence of inattention. Whereas the last subtype of ADHD is combined (ADHD-C) and exhibits all symptoms of inattention and hyperactivity [7]. ADHD patients show clear differences in the functional and structural parts of the brain compared to normal people. These differences particularly involve changes in neuronal connection between specific regions of the brain, mostly accompanied by alternations in the volume of the brain observed in neuroimaging [8], [9]. These anatomical variances have been associated with changes in a person's motivation regulation, cognitive function, and attention [10]. Moreover, in ADHD cases the reward system of the brain which primarily utilizes the neurotransmitter dopamine is affected [11]. For example, abnormalities in presynaptic storage have been seen in the prefrontal cortex of ADHD patients, critically impairing the person's cognitive process, attention, and working memory [12].

Fortunately, there is increasing evidence that these functional and structural alternations may not be static. These parts of the brain of ADHD patients can resemble the Typical Developmental Controls (TDCs) through timely detection and treatment of ADHD symptoms. For example, Mattfeld et al. [8] discovered that people who recovered from ADHD have restored normal brain connectivity during the period of rest as compared to the individuals still experiencing symptoms of ADHD. Effective management of ADHD can lead to the enhancement of quality of life and improved integration into society. Therefore, it is important to diagnose ADHD as early and as accurately as possible, and also perform evidence-based management so that the outcomes are long-term [5].

Currently, the most common method utilized for the diagnosis of ADHD is clinical one. It is based on the Diagnostic and Statistical Manual of Mental Disorders (DSM) and is published by American Psychiatric Association. Currently, its 5th edition is in use, named as DSM-5. In this method the clinician needs to look at the person's neurological status, for this purpose, they may go through the battery of examinations. The criteria utilized in this method are derived from the symptoms reported by the patients, or the subjective opinions of their spouse, parents, teacher, or any other family member. However, this is a subjective measure that may overlook the underlying neurobiological mechanisms [13]. Therefore, various studies have start utilizing neuroimaging modalities such as Magnetic Resonance Imaging (MRI) [14], physiological signals like electrocardiogram (ECG) [15] and electroencephalogram (EEG) [16], and other modalities like accelerometer [17]. These are more objective measures as

compared to clinical assessment and utilize Artificial Intelligence (AI) techniques such as Machine Learning (ML) and Deep Learning (DL) to diagnose ADHD more efficiently and cost-effectively.

However, the diagnosis of ADHD is quite an intensive area of research. Consequently, there is a dire need to evaluate various diagnostic tools and AI approaches utilized for the diagnosis of ADHD in the existing state-of-the-art research. In this regard, this study performed a Systematic Literature Review (SLR) in this field. To the best of our knowledge, there is only one article [20] that has conducted an SLR on diagnostic tools and AI approaches, comprising both ML and DL for diagnosing ADHD. This article reviewed 92 studies from 2010-2021. However, the AI has seen a rapid advancement in recent years that has also contributed to diagnosis of ADHD. Therefore, in this study, we conduct an SLR from 2021 onward and thoroughly analyze 98 research studies to find various diagnostic tools and AI techniques utilized in the current research for the identification of ADHD. The research questions which are intended to be answered are given below.

RQ1: Which Diagnostic tools or modalities are primarily utilized for the identification of ADHD and how effective are these modalities?

RQ2: What are the primarily employed ML and DL models for the identification of ADHD?

RQ3: What are the mainly utilized validation approaches to classify ADHD patients from TDCs?

RQ4: Which other comorbidities are considered with ADHD in current research?

RQ5: What are the gaps in the current research and which key improvements are required to diagnose ADHD more efficiently?

The key contributions of this article are mentioned below.

- Firstly, a detailed examination of recent trends and developments in the diagnosis of ADHD has been carried out. Particularly, the latest 98 studies from 2021 to 2024 are considered for this purpose, by utilizing the standard guidelines of SLR [18], [19].
- Secondly, the identification and analysis of various diagnostic tools, ML and DL techniques employed for ADHD diagnosis are performed. This analysis assists the researchers in selecting the right diagnostic tool and AI technique.
- Lastly, the significant research gaps where improvements have been required in ADHD diagnosis are highlighted in this study.

II. REVIEW PROTOCOL

There are seven basic elements in the review protocol. The first two elements background and research questions are already presented in the introduction section. Consequently, these two elements are excluded in this section. However, the description of the other five elements is presented in subsequent sections. The overview of the SLR is presented in Figure 1.

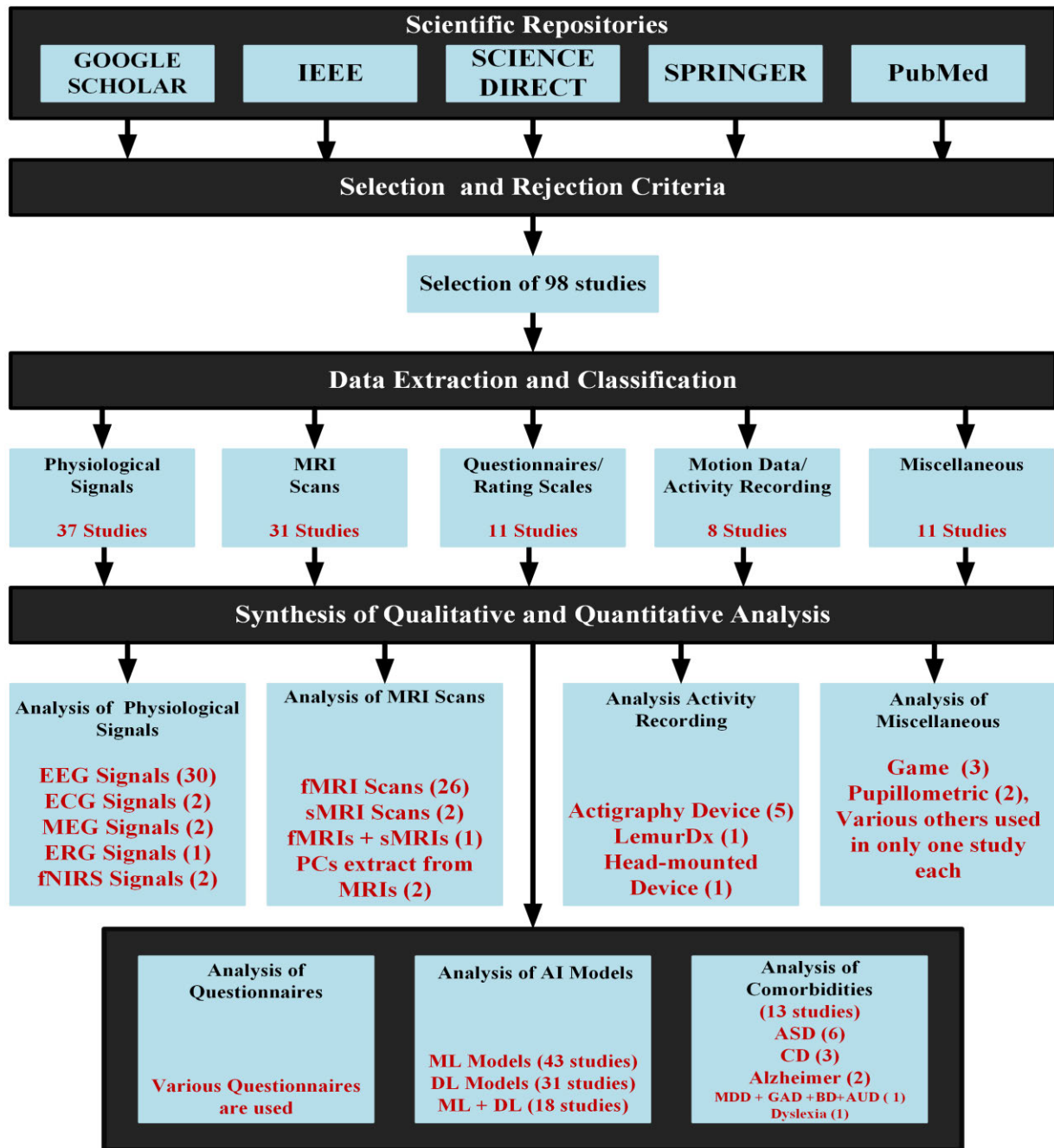


FIGURE 1. Overview diagram of systematic literature review.

A. CATEGORIES DEFINITION

To simplify the process of extracting and synthesizing data, we define five categories. The explanation of each category is given below.

1) PHYSIOLOGICAL SIGNALS

The research studies that utilize physiological signals as a diagnostic tool for ADHD diagnosis are placed in this category. These signals have become a powerful tool due to their enhanced objectivity and reliability. These are the responses

from both the central and peripheral nervous system. Examples include electroencephalogram (EEG), electrocardiogram (ECG) and electroretinogram (ERG) etc. [21]. Moreover, magnetoencephalography (MEG) is also a type of physiological signals, but instead of recording electrical activities like EEG, these signals record the magnetic activities of the brain [22]. Another type of physiological signals is functional near-infrared spectroscopy (fNIRS), which measures changes in hemodynamic signals reflecting neural activity [23].

2) MRI

Studies where MRI is employed as a diagnostic tool for the identification of ADHD fall under this category. MRI scans emerged as a powerful tool for understanding the structural and functional changes of the brain and are widely used by medical researchers for the diagnosis of ADHD. It comprises two types: Structural Magnetic Resonance Imaging (sMRI), which creates detailed images of brain structure and helps to identify anatomical changes in the brain [24]. Functional Magnetic Resonance Imaging (fMRI) appeared as a promising auxiliary diagnostic tool; it helps to study how different parts of the brain work together [25].

3) QUESTIONNAIRE/RATING SCALE

Research articles that utilize questionnaires or rating scales as a diagnostic tool for the detection of ADHD categories into this group. A list of printed or mimeographed questions is known as a questionnaire which is filled out by or for respondents to gather their opinions. The main objective of the questionnaire is to gather quantitative primary data [26]. Medical professionals utilize various rating scales or questionnaires to diagnose ADHD.

4) MOTION DATA/ ACTIVITY RECORDING

Research articles employed motion data or data recorded through activities as ADHD diagnostic tools grouped into this category. There are several devices such as accelerometer [17], and actigraphy [107] utilized to record activities.

B. OTHERS

From the overall selected 98 studies, the studies that utilized the least common modalities for ADHD diagnosis fall under this category. These modalities are utilized in one, two, or a maximum of three research studies.

C. INCLUSION AND EXCLUSION CRITERIA

Inclusion, as well as exclusion criteria, are defined to carry out this SLR for acquiring the desired objectives. For this purpose, the following specific rules are defined.

- First of all, the studies that are published in well-known repositories such as Google Scholar, IEEE, Science Direct, Springer, and PubMed are selected.
- The articles that contain the keywords ADHD diagnosis, ADHD detection, or ADHD identification in the title or abstract are included. Moreover, the content of the title and abstract of the selected studies should be relevant to the research questions.
- The journals and conference proceedings published from 2021 to 2024 are taken into consideration. All those articles that are published before 2021 are excluded to ensure inclusion of the latest articles.
- Chosen studies must be result-oriented and backed by clear evidence and experimentation. All other articles which do not demonstrate clear results are discarded.

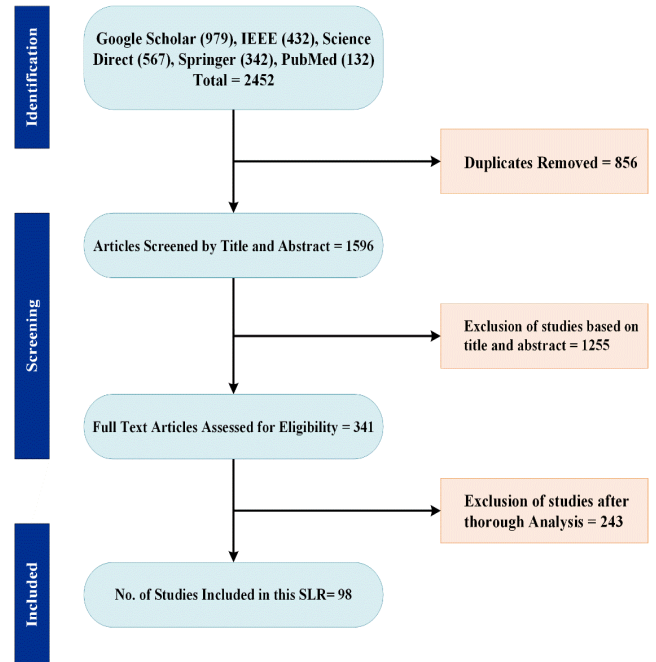


FIGURE 2. PRISMA Flow Chart for selecting and retrieving the studies.

- The studies that have almost similar content are excluded and only one of them is selected.

D. SEARCH PROCESS

In this study, the search process commences by utilizing five repositories (PubMed, IEEE, Google Scholar, Springer, and Science Direct) as mentioned in the selection and rejection criteria. Various keywords or search terms have been utilized during the search process; its overall summary is presented in TABLE 1. To conduct this process two types of operators such as, AND, and OR are employed. The outcomes obtained using the AND operator are not enough which necessitates the utilization of the OR operator. Nevertheless, the outcomes collected from OR operators are exceedingly vast, so scanning all of these outcomes is not feasible. Consequently, advanced search options are taken into consideration, offered by chosen repositories e.g. where the keyword comprises “time span” for getting the more accurate results. After the evaluation of initial findings, we opt to select only those studies which highly related to ADHD diagnosis using ML or DL.

Figure 2 presents the PRISMA flow chart illustrating the inclusion strategy for the retrieved papers.

- Initially, 2452 studies are evaluated and 856 duplicate articles are removed.
- Consequently, the remaining 1596 research articles are considered and 1255 articles are excluded after reading their title and abstract.
- We thoroughly evaluate the remaining 341 studies. Based on this evaluation 243 research studies are further

TABLE 1. Summary of search keywords with outcomes.

Sr. #	Search Keyword	Operator	Google Scholar	Science Direct	IEEE	PubMed	Springer
1	ADHD Detection/Diagnosis/ Identification using ML	N/A	40100	1670	95	315	1814
	ADHD Detection/Diagnosis/ Identification using DL	N/A	42400	3182	55	91	2501
2	ADHD ML, MRI	AND	2680	802	23	109	844
		OR	61500	601289	21875	78102	10000
3	ADHD DL, MRI	AND	2840	1388	17	38	1121
		OR	78400	602219	21821	779979	10000
4	ADHD ML, Physiological Signals	AND	2440	673	3	3	527
		OR	29400	100000+	12818	167657	10000
5	ADHD DL, Physiological Signals	AND	2570	1174	2	3	668
		OR	33200	100000+	12762	167433	10000
6	ADHD ML, Questionnaires/ Rating scale	AND	2990	640	5	24	767
		OR	63500	100000+	19733	1621339	10000
7	ADHD DL, Questionnaires/ Rating scale	AND	3800	1175	4	9	9391
		OR	78400	100000+	19677	1621130	10000
8	ADHD ML, Motion data/activity recording	AND	2710	532	4	12	539
		OR	28000	717307	19787	300589	10000
10	ADHD DL, Motion data/activity recording	AND	3150	932	2	5	717
		OR	35600	718422	19732	300372	10000

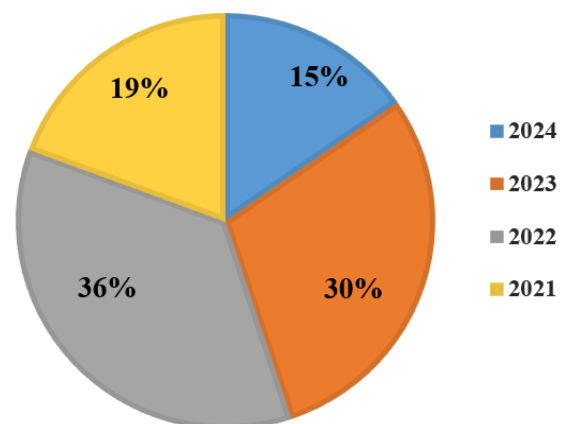
discarded and 98 studies that completely adhere to our selection and rejection criteria are included.

E. QUALITY ASSESSMENT

To ensure the credibility of this SLR, we have chosen high-impact studies, such as those obtained from reputable databases that are authentic and accepted internationally. TABLE 2. illustrates our efforts to select high-impact studies and also presents a comprehensive overview of the repositories w.r.t their publication types. **Repository** represents the names of research repositories from where studies have been selected. **Type** indicates that the chosen study either belongs to a journal or conference proceeding. **References** are provided for the selected studies. Lastly, the **Total** indicates the overall number of journal or conference papers belongs to each repository.

From TABLE 2 it can be observed that forty-seven (47) studies have been selected from Google Scholar, including forty-six (46) journals and one (1) conference, twenty (20) studies from IEEE, comprising ten (10) journals and ten (10) conferences.

Similarly, nineteen (19) research studies have been chosen from Springer, encompassing eighteen (18) journals and one (1) conference. Moreover, eleven (11) studies selected from Science Direct and one (1) study from PubMed, studies included from both these databases are all journals. The overall investigation shows that out of 98 selected research studies, 86 studies are published in journals, while 12 presented at international conferences. This indicates that 88% of

PUBLICATIONS PER YEAR**FIGURE 3.** Distribution of studies w.r.t per year.

our literature published in journals while the remaining 12% presented at conferences.

Additionally, the objective is also to examine the state-of-the-art and latest research studies. Therefore, the distribution of studies w.r.t their publication year is depicted in Figure 3. Specifically, fifteen (15) research studies have been considered from 2024, twenty-nine (29) studies from 2023, thirty-five (35) studies from 2022, and nineteen (19) studies from 2021.

F. DATA EXTRACTION AND SYNTHESIS

A complete template to extract data as well as perform synthesis has been developed and presented in TABLE 3. The initial

TABLE 2. Summary of selected studies according to scientific repositories and publication type.

Repository	Type	Reference	Total
PubMed	Journal	[63]	1
	Conference	N/A	
IEEE	Journal	[32][26][84][85][86][87][88][89][120] [64]	20
	Conference	[35][46][53][61][69][77][79][92][110][122]	
Google Scholar	Journal	[28][29][30][36][37][38][39][41][42][43][45][47][51][52][57][59][60][67][70][76][80][81][82][83][90][93]	47
	Conference	[96][98][99][101][102][103][104][105][107][108][109][112][113][114][115][116][117][118][119][123]	
Springer	Journal	[34][44][48][49][58][62][66][68][72][73][78][91][94][95][97][100][111][121]	19
	Conference	[54]	
Science Direct	Journal	[31][33][40][55][56][71][74][75][25][106][124]	11
	Conference	N/A	

step involves the extraction of bibliographic information such as title, publication year, and type of research study. Afterward, essential findings such as proposed methodologies, implementation details, and experimentation process of each included study have been extracted. This extracted data helps to carry out the detailed examination to achieve the objectives of this SLR. Lastly, a thorough analysis is conducted to get the answers to the research questions outlined in Section I.

III. RESULTS AND ANALYSIS

The selected studies are divided into five major categories as illustrated in TABLE 4. Additionally, the references of subsequent studies are also provided. It can be seen in TABLE 4 that thirty-seven (37) research studies fall in the category of physiological signals, thirty-one (31) studies belong to the MRI category, eleven (11) research studies are associated with a questionnaire or rating scale diagnostic tool, while eight (8) studies placed into motion data or activity recording category. Lastly, eleven 11 studies fall under the others category.

A comprehensive analysis of each category is carried out to ensure accurate outcomes can be presented and the goal of SLR is fulfilled. Their detail is provided in subsequent sections where we evaluate them based on certain parameters.

A. PHYSIOLOGICAL SIGNALS

Several studies utilize physiological signals as a diagnostic tool for the identification of ADHD. These signals coming from both the peripheral and central nervous system, have emerged as useful tools thanks to their objectivity and dependability. TABLE 5 indicates the analysis of ADHD diagnosis based on various types of physiological signals with parameters defined below. Firstly, **Signal Type** specifies the type of physiological signals employed for the diagnosis of ADHD. Secondly, **Dataset** denotes the size of data used in the study, while **Age** specifies the age of participants from whom the data have been collected. **Features** indicate the type of features extracted from signals and utilized as input to the classification models and **ML/DL Model** indicates that which model is utilized for classification or detection

of ADHD. Furthermore, the **Validation Approach** indicates that the split ratio used for training and testing purposes. Lastly, the **Accuracy %** shows the results obtained in each study in terms of percentage of accuracy. The summary of the studies that utilized physiological signals is presented in the subsequent paragraphs.

In TABLE 5. Alim and Imtiaz [27] developed an EEG signals-based algorithm for the identification of ADHD. They collected the EEG signals from a publicly available database known as, IEEE data port.¹ Subsequently, they employed the first four bands to extract statistical, time domain, and frequency domain features followed by feature selection using Principal Component Analysis (PCA). Gaussian Support Vector Machine (SVM) based model is utilized for classification using the holdout method with a 70:15:15 (train-valid-test) split ratio and 10-fold Cross-Validation (CV), resulting in a maximum accuracy of 94%. Ahire et al. [28] discussed the utilization of machine learning algorithms and EEG-based data for the classification of ADHD. For this purpose, they employed data from the IEEE data port, features are extracted from the time domain, frequency domain, and time-frequency domain. Four popular classifiers such as K Nearest Neighbor (KNN), Random Forest (RF), Logistic Regression (LR), and Extreme Gradient Boosting (EGB) are utilized for classification and a maximum of 87% accuracy is achieved with KNN. In another study [29] they analyzed morphological and frequency domain (Power Spectral Density) features from the same EEG data and PCA has been used for dimensionality reduction. For classification four classifiers are employed: AdaBoost, KNN, NB, and RF, and 96% enhanced accuracy is achieved. Moreover, Loh et al. [30] presented a study which discussed the development of Deep Neural Network (DNN) for the detection of ADHD and Conduct Disorder (CD) based on ECG signals. They obtained this dataset from Child Guidance Clinic, Singapore, which contains data from 123 participants, including 45 ADHD patients 16 CD and 62 incorporate both with age ranges between 7 to 16 years. Features are extracted and

¹<https://ieee-dataport.org/open-access/eeg-data-adhd-control-children>

TABLE 3. Data extraction and synthesis template.

Sr.#	Description	Details
1	Bibliographic Information	The title, year of publication, and research paper type (such as journal or conference) is examined.
2	Proposed Methodology	The methodology proposed in each selected study is observed.
3	Implementation Details	Technologies utilized in each selected study to implement the proposed methodology are examined.
4	Results	A thorough analysis has been carried out to analyze the results of each selected study.
5	Categorization	Classification of selected research papers is performed according to categories defined in Section III. The summary of the results is presented in TABLE 4.
6	Analysis of Categories	To find the answers to research questions investigation of each category is carried out. The results are summarized as follows. Physiological Signals (TABLE 5), MRI (TABLE 6), Questionnaire\ Rating Scale (TABLE 8), Motion Data\ Activity Recording (TABLE 9), Others (TABLE 10).

TABLE 4. Classification results of selected studies.

Sr.#	Category	References of Corresponding Studies	Total
1	Physiological Signals	[28][29][30][31][32][33][34][35][36][37][38][39][40][41][42][43][44][45][46][47][48][49][50][51][52][53][54][55][56][57][58][59][60][61][62][63][64]	37
2	MRI Scans	[66][67][68][69][70][71][72][73][74][75][76][77][78][79][80][81][25][26][82][83][84][85][86][87][88][89][90][91][92][93][94]	31
3	Questionnaires/Rating Scales	[95][96][97][98][99][100][101][102][103][104][105]	11
4	Motion Data\ Activity Recording	[106][107][108][109][110][111][112][113]	8
5	Others	[114][115][116][117][118][119][120][121][122][123][124]	11

classification is carried out using 1D Convolutional Neural Network (CNN) with 10-fold CV. Moreover, Gradient-weighted Class Activation Mapping (Grad-CAM) is also implemented to provide time localization of ECG features, which helps for diagnosis. The result shows that the proposed model outperforms as compared to others with 96.04% accuracy. Maniruzzaman et al. [31] proposed an approach for the detection of ADHD, employing optimal channels and features from EEG signals. They sourced the data from the IEEE data port, time domain, morphological, and nonlinear features are extracted and selected from six channels using Least Absolute Shrinkage and Selection Operator (LASSO). Classification is performed using six classifiers as Gaussian Process Classification (GPC), RF, KNN, MLP, DT, and LR, with 5-fold CV, resulting in a maximum 97.53% impressive accuracy. Khare and Acharya [32] proposed an explainable and interpretable model for the detection of ADHD. The methodology involves utilizing EEG signals obtained from the IEEE data port. Additionally, the combination of Variational Mode Decomposition (VDM) and Hilbert Transform (HT) which is (VDM-HT) is employed to extract statistical, entropy-based, and nonlinear features. Lastly, for making decisions and interpretation the Explainable Boosted Machine (EBM) model integrated with black box explainers like LIME, SHAP, PDP, and Morris are used with a 10-fold CV. The results reveal that the proposed model gives excellent results with 91.81 accuracy. Furthermore, Alves et al. [33]

utilized Event-related potential (ERP) microstate networks based on EEG signals for the classification of AD HD. The methodology involves collecting ERP signals from a public database, which consists of 144 children's data, including 44 TDCs, 52 ADHD-I, and 48 ADHD-C aged between 11 to 12 years. Subsequently, the topological features are extracted from ERP microstate networks, and classification is carried out using a Neural Network (NN) with Leave-one-out Cross Validation (LOOCV), achieving a very good 99.72% accuracy.

Trinh et al. [34] investigated EEG signals data to classify ADHD using machine learning. For this purpose, they employed two types of datasets, the task-based EEG signals consist of 122 participants having 24 ADHD patients and 78 TDCs, and the resting-state EEG signals contain data from 150 subjects, including 52 ADHD patients and 98 TDCs. Subsequently, the spectral power is extracted (frequency based features) and classification is carried out using four classifiers such as, SVM, KNN, RF, and Elastic Net with nested CV. The experimental results reveal that the achieved Area Under the Curve (AUC) score is 64.2 with task-based signals. Swadi and Miften [35] discussed the utilization of machine learning and EEG signals for the diagnosis of ADHD. The methodology involves collecting data from the IEEE data port. Moreover, statistical features are extracted from frequency bands using a Discrete Wavelet Transform (DWT), followed by feature selection using

TABLE 5. Summary of studies that utilize Physiological signals for the diagnosis of ADHD.

Sr.#	Ref.	Signal Type	Dataset	Age	Features	ML/DL Model	Validation Method	Accuracy %
1	[28]	EEG	60 ADHD, 60 TDCs	7 to 12	Statistical, Time & Frequency	Gaussian SVM, PCA	Holdout (70:15:15), 10-fold CV	94
2	[29]	EEG	61 ADHD, 60 TDCs	7 to 12	Time, Frequency, Time-frequency	LR, RF, EGB, KNN	N/A	87
3	[30]	EEG	61 ADHD, 60 TDCs	7 to 12	Morphological and Frequency domain	AdaBoost, KNN, NB, RF, PCA	N/A	96
4	[31]	ECG	45 ADHD, 62ADHD+CD, 16 CD	7 to 16	N/A	CNN with Grid-CAM	10-fold CV	96.04
5	[32]	EEG	61 ADHD, 60 TDCs	7 to 12	Time domain, morphological, non-linear	(GPC), RF, KNN, , MLP, DT, LR, LASSO	5-fold CV	97.53
6	[33]	EEG	61 ADHD, 60 TDCs	7 to 12	Entropy-based, statistical, nonlinear	EBM, (LIME, SHAP, PDP, Morris)	10-fold CV	.91.81,
7	[34]	ERP data (EEG based)	44 TDCs, 52 ADHD-I, 48 ADHD-C	11 to 12	Topological	NN	LOOCV	99.72
8	[35]	EEG	52 ADHD, 98 TDCs, 24 ADHD, 78 TDCs	N/A	Frequency domain	SVM, KNN, RF, Elastic Net	nested CV	AOC. 64.2
9	[36]	EEG	60 ADHD, 61 TDCs	7 to 12	Frequency domain	least-square SVM, KNN, DT, NB, SVM, K-mean clustering	10-fold CV	96.49
10	[37]	EEG	31 ADHD, 30 TDCs	7 to 12	Frequency domain	CNN with LRP	5-fold CV	94.52
11	[38]	EEG	61 ADHD, 60 TDCs	7 to 12	19 EEG channels	SVM, RF, NB, DT, AdaBoost, LDA	CV	84
12	[39]	EEG	5 ADHD, 5 TDCs	7 to 12	Entropy-based	ANN	CV	99.82
13	[40]	EEG	61 ADHD, 60 TDCs	7 to 12	Time domain, Frequency domain	CNN, SVM, LR, RF	5-fold	95.83
14	[41]	EEG	61 ADHD, 60 TDCs	7 to 12	morphological, Time-domain	SVM, MLP, KNN, LR, LASSO	LOOCV	95.83
15	[42]	EEG	61 ADHD, 60 TDCs	7 to 12	Textual, Statistical	KNN	10-fold CV, LOOCV	97.19%
16	[43]	EEG	1843 ADHD, 2330 TDCs	N/A	Time domain, Frequency domain	Weighted KNN, IHMV, NCA	10-fold CV, LOOCV	95.57
17	[44]	ERP (EEG based)	30 ADHD, 30 TDCs	4 to 15	Frequency domain	NB, GLM, LR, DL, DT, RF, SVM	10-fold CV	99.85
18	[45]	EEG	30 ADHD, 30 TDCs	55-102 months	Time domain	LSTM	10-fold CV, LOOCV	90.50
19	[46]	EEG	77 ADHD, 80 TDCs	N/A	N/A	NB, LR, KNN, PCA		86
20	[47]	EEG	5 ADHD, 5 TDCs	7 to 12	Higuchi and Katz nonlinear	ANN, KNN, SVM, LD, LR	10-fold CV	100
21	[48]	EEG	61 ADHD, 60 TDCs	7 to 12	N/A	CNN	5-fold CV	100
22	[49]	EEG	8 ADHD, 8TDCs	7 to 18	Frequency domain	SVM, LSTM, FFBPNN	80: 20	92.15
23	[50]	EEG	ADHD 61, TDCs 60	7 to 12	Nonlinear (Higuchi, Katz and Petrosian)	Ensamble learning, CFS, LASSO, Logistic, RFE	80:20	98.33
24	[51]	EEG	ADHD 60, TDCs 60	7 to 12	N/A	AdaBoost, RF, SVM	N/A	84
25	[52]	EEG	34 ADHD, 45 TDC	18 to 45	MMN	Linear SVM, Fisher score	LOOCV	81.01
26	[53]	EEG	30 ADHD, 37 TDCs	N/A	time series	HMMs, SVM	5-fold CV	90.0
27	[54]	EGG	61 ADHD, 60 TDCs	7 to 12	Frequency domain	RF, KNN, SVM, ANN,	5-fold CV	81.82
28	[55]	ECG	45ADHD, 62ADHD+CD, 16 CD	7 to 16	Entropy	DT, KNN, AdaBoostM2, Bagged Tree,Subspace, SVM, SFS	10-fold CV	87.19
29	[56]	EEG	45ADHD, 62ADHD+CD, 16 CD	7 to 16	Nonlinear	DT, KNN, AdaBoostM2, Bagged Tree,Subspace, SVM, SFS	10-fold CV	97.88

TABLE 5. (Continued.) Summary of studies that utilize Physiological signals for the diagnosis of ADHD.

30	[57]	EEG	61 ADHD 60 TDCs,	7 to 12	N/A	EEG-MHCNet	10 fold CV	82.81
31	[58]	EEG	61 ADHD, 60 TDCs	7 to 12	Images	CNN, ResNet	10-fold CV	96.8
32	[59]	EEG	61 ADHD, 60 TDCs	7 to 12	Statistical	NB, RF, DT, KNN, SVM, AdaBoost, LDA	5-fold CV	84
33	[60]	eyes-open rs- MEG	25 ADHD, 25 TDCs	18 to 23	GC	Ensemble Learning, mRMR	LOOCV	89.82
34	[61]	rs-MEG	25 ADHD, 25 TDCs	18 to 23	COH	SVM with RBF, KNN, RF, NCA	LOOCV	92.7
35	[62]	ERG	94 ASD, 46 ADHD, 146 TDCs	11 to 15	Time domain, Time- frequency domain	RF, AdaBoost, GradBoost, XGBoost, SVM, KNN, MLP	10-fold CV	70
36	[63]	Task-based fNIRS	33 ADHD 39 TDCs	8 to 15	Time domain	RLDA	LOOCV	82
37	[64]	Task-based fNIRS	72 ADHD, 171 TDCs	6 to 13	fNIRs and behavioral	SVM, KNN, LR, NN, LDA, RF	10-fold CV	91.67

k-means clustering. Lastly, classification is carried out by employing five well-known classifiers such as SVM, least Square SVM, KNN, RF, and elastic Net with 10-fold CV. The results indicate that the maximum achieved accuracy is 96.49% with Least-square SVM. Nouri et al. [36] introduced a deep learning model based on EEG signals for the diagnosis of ADHD. The dataset utilized in this study is the “First National EEG Data Analysis Competition with Clinical Application” which consists of 61 participants, which include 31 ADHD patients and 30 TDCs aged between 7 to 12 years. The frequency domain features are extracted and Layer-wise Relevance Propagation (LRP) is employed for feature selection and visualization. Moreover, CNN is used to classify the data with 5-fold CV, resulting in an accuracy of 94.52%. Besides this, Chauhan and Choi [37] suggested that combinations of regional data increase the classification accuracy for the diagnosis of ADHD. They utilized EEG data collected from the IEEE data port, containing 61 ADHD and 60 TDCs with ages between 7 to 12 years. The data collected from 19 channels used as input features to the five well-known classifiers, including Random Forest (RF), Support Vector Machine (SVM), Naïve Bayes (NB), Decision Tree (DT), Linear Discernment Analysis (LDA), and AdaBoost with CV. The results show that this approach can achieve 84% maximum accuracy with NB.

Catherine et al. [38] proposed a study for the identification of ADHD based on EEG signals using NN and entropy measures. The data is obtained from 10 subjects, including 5 from each category: ADHD and TDCs, aged 7 to 12 years. Entropy-based features are extracted and classification is performed by NN with CV, which results in a maximum of 99.82% an impressive accuracy. TaghiBeyglou et al. [39] presented a novel CCN structure with a classical model based on raw EEG data for ADHD detection. The dataset is collected from the IEEE data port, time domain and frequency domain features are extracted using CNN and classification is carried out with three popular classical classifiers such as

SVM, LR and RF with 5-fold CV, resulting in an accuracy of 95.83%. Maniruzzaman et al. [40] introduced an approach using feature selection and machine learning models for ADHD detection. The dataset utilized in this study is EEG signals collected from the IEEE data port, the morphological and time domain statistical features are extracted followed by feature selection using LASSO and t-test. Subsequently, classification is carried out using four classifiers, Multi-Layer Perceptron (MLP), KNN, SVM and LR with LOOCV. The results indicate that the maximum accuracy achieved in this study is 94.2%. Tanko et al. [41] proposed EPSPatNet86, a novel hand-modeled learning network for the detection of ADHD. This model utilized EEG signals collected from IEEE data port, the textual and statistical features are generated using EPSPat and statistical methods followed by feature selection using IChi2. Subsequently, KNN is utilized for classification with 10-fold CV and LOOCV, resulting in an impressive accuracy of 97.19%. Similarly, Barura et al. [42] also proposed a hand-modeled EEG classification model namely, TMP19 with a new ternary motif pattern (TMP). The dataset utilized in this study consists of 4000 EEG signals, of which 2330 belong to TDCs and 1843 belong to signals of ADHD patients. Statistical features and motifs have been extracted using TMP and statistics in both time and frequency domains, followed by feature selection using Neighborhood Component Analysis (NCA). Subsequently, KNN is used for classification, and for enhanced accuracy, Iterative Hard Majority Voting (IHMV) is utilized with both 10-fold CV and LOOCV. The proposed approach resulted in maximum accuracy of 95.57%. Ghasemi et al. [43] effectively utilized machine learning models for detecting ADHD based on Event-Related Potential (ERP) signals derived from EEG. The dataset comprises 60 participants with 30 belonging to each category (ADHD and TDCs) aged between 4 to 15 years. Frequency domain features are extracted, and classification is performed with seven classifiers such as Generalized Linear Model (GLM), NB, LR, DL, DT, RF and SVM,

selecting specific patterns for each model using threshold. 10-fold CV is employed for training and testing purposes and a very good 99.85% accuracy is achieved. Moreover, Chang et al. [44] presented a Long Short-Term Memory (LSTM) network based on EEG signals for discriminating ADHD patients from TDCs. The dataset consists of 60 subjects, including 30 from each category with age ranges between 55 to 102 months. Temporal information is extracted from input sequence and classification is performed by LSTM using 10-fold CV and LOOCV. The results show that the proposed model achieved a maximum of 90.50% accuracy. Saini et al. [45] employed machine learning models for the classification of ADHD using EEG data. The dataset is contained of 157 children out of which 80 are TDCs and 77 are ADHD patients. Features are extracted from EEG signals followed by feature selection using PCA. Subsequently, NB, LR, and KNN are utilized for classification, resulting in an accuracy of 86%.

Joy et al. [46] presented an approach for the detection of ADHD using Tunable Q Factor Wavelet Transform (TQWT). The dataset consists of ADHD signals collected from 10 children, 5 belonging to each category: ADHD and TDCs, aged between 7 to 12 years. Features are extracted using TQWT and evaluated using Higuchi and Katz nonlinear fraction dimensional methods. Afterwards, these features are fed as input to various classifiers such as ANN, KNN, SVM, LD and LR for classification with 10-fold CV. The maximum accuracy achieved in this study is 100% with ANN. Additionally, Hassan and Singhal [47] introduced a lightweight CNN model for ADHD diagnosis using EEG signals obtained from IEEE data port. After preprocessing Canonical Correlation Analysis (CCA) is employed to maximize the correlation between signals, which then passed to proposed model for classification using 5-fold CV. The results indicate that the proposed model achieves maximum accuracy of 100%. Tosun [48] focused on detecting the most effective channels and recording the status of the diagnosis of ADHD utilizing EEG signals. The dataset employed in this study comprises data from 16 subjects, with 8 from each category: ADHD and TDCs, aged between 7 to 18 years. The spectral features (frequency-based) are extracted, and classification is carried out using Feedforward Backpropagation Neural Network (FFBNN), LSTM and SVM. 80: 20 ratio utilized for training and testing purposes, resulting in an accuracy of 92.15%. Another method for the detection of ADHD is proposed by Huong et al. [49] using non-linear features of EEG signals. The methodology involves collecting EEG data from IEEE data port, three nonlinear Fractal Dimensions (FD) features, such as Higuchi, Petrosian and Kutz are calculated followed by feature selection using various methods including, Filter method, Correlation-based Feature Selection (CFS), LASSO, Logistic, Recursive Feature Elimination (RFE) and Wrapper method. Subsequently, classification is carried out with ensemble learning comprising, Boosted Trees, Bagged Trees, Subspace KNN, Subspace Discriminant, Random Under-Sampling (RUS), and Boosted Tree. For training and testing

purpose 80:20 ratio split is employed, resulting in a maximum of 98.33% impressive accuracy. Parashar et al. [50] presented a study using machine learning models for ADHD diagnosis based on EEG data. The dataset employed from the IEEE data port, EEG signals from 19 channels either individually or in combination employed as input features for classification. Three well-known classifiers AdaBoost, SVM, and RF used for classification, resulting in an accuracy of 84%. In addition, Kim et al. [51] investigated Mismatch negativity (MMN) features as biomarkers to identify ADHD in drug naïve adults. This study utilized the EEG dataset encompasses the data of 79 subjects, including 34 ADHD patients and 45 TDCs aged between 18 to 45 years. MMN features are extracted followed by feature selection using fisher score. Subsequently, linear SVM is employed for classification and its accuracy is evaluated using LOOCV. The results indicate that the proposed method achieves 82.35% accuracy.

Another approach is introduced by Maya-Piedrahita et al. [52] for ADHD detection using Hidden Markov Models (HMMs) and Probability Product Kernels (PPKs). The EEG dataset consists of 67 subjects, which include 30 ADHD patients, and 37 TDCs are employed. Time series features are extracted, using a set of trained HMMs and SVM fed by PPKs performed classification with 5-fold CV. The proposed approach resulted in a maximum accuracy of 90%. Moreover, Holker et al. [53] presented a framework for detecting ADHD using Quantitative EEG (QEEG) features. The dataset obtained from the IEEE data port. The QEEG includes spectral, amplitude, Functional Connectivity (FC), and range EEG features extracted from five frequency bands followed by feature selection using ANOVA, Chi-square, Gini Index, and Information Gain. Subsequently, four classifiers: RF, KNN, SVM, and ANN are utilized to discriminate TDCs and ADHD patients with a 5-fold CV. The experimental results reveal that this approach outperforms as compared to others with 81.82% accuracy. Koh et al. [15] developed an expert system for automatic classification of ADHD and CD using ECG signals. For this purpose, they employed ECG data from 123 subjects, including 45 ADHD patients, 16 CD, and 62 incorporating both ADHD and CD aged between 7 to 16 years. Entropy features are extracted from the modes of transformed ECG signals data, while ANOVA is utilized to determine the variability between classes. Subsequently, the discriminant features are fed to the six classifiers: DT, KNN, AdaBoostM2, Bagged Tree, Subspace, and SVM for classification. Training and testing are performed by 10-fold CV where in every fold Sequential Forward Selection (SFS) and ADASYN are employed to facilitate learning. The results show that the system developed in this study achieves a maximum accuracy of 87.19% with Bagged Tree. Similarly, Tor et al. [54] presented an automated system using EEG signals for the classification of ADHD and CD. The dataset consists of 123 participants, of which 45 belong to ADHD, 16 to CD and 62 showing symptoms of both, aged between 7 to 16. Nonlinear features are extracted from transformed EEG signals and ADASYN is employed to balance the data.

Subsequently, discriminant features are selected using SFS and fed to the array of classifiers, including DT, KNN, AdaBoostM2, Bagged Tree, Subspace, and SVM. Training and testing are performed using a 10-fold CV, achieving an accuracy of 97.88%. Furthermore, Sanchis et al. [55] presented a study with the aim of identifying the brain regions that efficiently classify ADHD and TDCs using EEG signals. The dataset is collected from the IEEE data port, features are extracted from EEG data, while selected using forward and backward stepwise feature selection methods. Subsequently, these selected features fed as input to the custom model namely, EEG Multithread Convolutional Based Neural Network (EEG-MHCNet) for classification, using 10-fold CV. The study achieved an accuracy of 82.81%. An efficient approach for the diagnosis of ADHD is proposed by Jahani and Safaei [56] using EEG signals. These signals are collected from the IEEE data port, then converted into RGB images, and passed as input to the two popular models known as CNN and Residual Network (ResNet), which extract features and perform classification. The performance of these models is evaluated using a 10-fold CV and results indicate the maximum accuracy (96.8%) is achieved with ResNet. Deshmukh et al. [57] explore brain regions' contributions to ML-based classification of ADHD using EEG signals. They sourced the signals from the IEEE data port, extracted statistical features, and performed classification using six well-known classifiers: NB, RF, DT, KNN, SVM, Adaboost, and LDA with a 5-fold CV. The maximum achieved accuracy is 84% with RF. Hamdi et al. [58] introduced a connectomics approach for detecting ADHD using eyes-open resting-state MEG signals. These signals obtained from Open MEG Archive,² a free MEG dataset is provided, consisting data from 50 subjects, including 25 from each category: ADHD and TDCs, aged between 18 to 23 years. Afterward, by utilizing Granger causality (GC), they calculated connectivity between MEG sensors, while the most discriminant measures selected by mRMR, which then utilized as input features to classifiers. For classification ensemble learning techniques including DT, SVM with RBF, KNN and NB is employed using LOOCV, achieving an accuracy of 89.82%. Similarly, in another study [59] they calculated Coherence (COH) between the Meg sensors in conventional frequency bands and then most discriminant COH measures have been selected by NCA. The selected measures utilized as input to three classifiers: SVM, KNN, and RBF for classification using LOOCV, resulting in an enhanced accuracy of 92.7%. Besides this, Manjur et al. [60] explored the potential of ERG signals for the diagnosis of ADHD and Autism Spectrum Disorder (ASD). The ERG signals obtained from 286 participants, which include 46 ADHD, 94 ASD and 146 TDCs with age ranges between 11 to 15 years. Subsequently, time domain, and time-frequency domain (DWT & VFCDM) features are extracted followed by the optimal features fed as input to the classifiers such as RF, AdaBoost, GradBoost,

XGBoost, SVM, KNN and MLP for classification. For validation 10-fold CV is employed, resulting in a maximum of 70% accuracy. Yang et al. [61] proposed a machine learning-based approach for the diagnosis of ADHD based on prefrontal functional near-infrared spectroscopy. For this purpose, they employed task-based fNIRS data collected from 72 participants during the Stroop task analysis. In these participants, 33 are ADHD patients and 39 are TDCs, with age ranges between 8 to 15 years. Consequently, time domain features are extracted and, then important features are selected and classified by using regularized LDA with LOOCV. The experimental results reveal that the proposed model achieves 82% accuracy.

From TABLE 5, it is analyzed that physiological signals are the most widely used modality for automatic diagnosis of ADHD, with 37 out of 98 studies utilizing this modality. Among these studies, ML techniques dominated a larger portion as compared to DL, where 21 out of 37 employed ML techniques, 8 utilized DL approaches, and 8 incorporated both ML and DL approaches. Moreover, EEG signals occupied the larger part among all physiological signals, with 30 studies out of 37 utilizing EEG signals, 2 used ECG signals, 2 employed rs-MEG signals and task-based fNIRS each, while ERG signals are utilized in 1 study. In terms of datasets, most of the studies that utilized EEG data obtained their data from a publicly available database known as IEEE data port, where 17 out of 30 studies utilized data from this database. This dataset consists of 121 participants, including 61 ADHD patients and 60 TDCs, aged between 7 to 12 years. Furthermore, frequency domain features emerged as the most commonly used features for the diagnosis of ADHD in studies that fall under this category, with 11 out of 37 utilized these features. To extract the frequency domain features the EEG signals are decomposed into five frequency bands known as delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and lower gamma (30-63 Hz) [27], [29], [58]. These frequency bands demonstrate significant differences between ADHD patients and TDCs. As Alim et al. [27] discovered that ADHD patients have a higher theta-to-beta ratio, while Kamida et al. [63] investigated that apart from the occipital region of the brain, ADHD patients exhibit higher beta activity in all other brain regions.

B. MRI

Several researchers employed MRI scans as a diagnostic tool for the diagnosis of ADHD. These scans become a potent tool to analyze the structural and functional changes in the brain and, have two types: Structural MRIs (sMRIs) and Functional MRIs (fMRIs). In TABLE 6 the analysis of ADHD diagnosis based on MRI scan is carried out according to the parameters defined below. Initially, **MRI Scans Type** indicates that MRIs (sMRI or\and fMRI) utilized for the diagnosis of ADHD. Next, **Dataset** shows the size of the dataset used for diagnosis. **Age** indicates the age range of the participants from whom data have been obtained. Afterwards,

²<https://www.mcgill.ca/bic/neuroinformatics/omega>

Features refer to the type of extracted attributes from the data that are feed into the classification model. Whereas **ML/DL Model**, specifies the type of model employed for ADHD detection, whether it's ML, DL, or both. The **validation Approach** specifies how the results are validated. Lastly, **Accuracy %** indicates the maximum accuracy achieved by any approach in percentage. The summary of these research studies is presented in the subsequent paragraphs.

In TABLE 6, Saurabh and Gupta [64] proposed a deep learning-based modified bidirectional LSTM (BLSTM) model for the classification of ADHD. For this purpose, they employed resting-state(rs)-fMRI data which consists of a total of 40 subjects, with 20 having ADHD and 20 TDCs, and extracted Resting State Network (RSN) features. Afterward, the data is split into the ratio of 80:20 for training and testing purposes, classification is performed with the proposed model, resulting in an accuracy of 87.50%. Shokair and Shalaby [65] presented an approach for the classification of ADHD using rs-fMRI data. They collected this data from the ADHD-200 dataset, comprising data of 1200 subjects, which include 600 with each category (ADHD and TDCs), aged between 7 to 26 years. Spatio-temporal features are extracted from images, and classification is carried out using CNN with various optimizers such as ADAM, SGDM, and RMS prop. A 70:30 split ratio is utilized for training and testing purposes, resulting in a maximum accuracy of 97.78 %. Sharma et al. [66] introduced a framework for the automatic diagnosis of ADHD based on machine learning. This framework utilizes rs-fMRI data along with its relevant phenotypic/personal characteristics (PC), data collected from the ADHD-200 dataset, which includes 400 subjects aged between 7 to 18 years. PC and Functional Connectivity (FC) features are extracted from the data, and Independent Component Analysis (ICA) is performed on FC features. Subsequently, data is split into 70:30 ratio, and classification is performed through an ensemble approach integrating DNN and SVM models. The experimental results reveal that this framework achieves 99% accuracy. Moreover, Balasubramanian et al. [67] proposed a machine-learning model for detecting and predicting ADHD. They employed rs-fMRI scans obtained from the public ADHD-200 database, which contains 328 subjects' data aged between 7 to 21 years. Temporal features are extracted from the data and the data split ratio 55:45 is applied for training and validation purposes. Data augmentation is carried out by 3DGAN and multi-model 3DCNN is employed to classify ADHD patients and TDCs. The results show that the proposed model achieves an accuracy of 97%. Salman et al. [68] performed classification between ADHD patients and TDCs by utilizing the Kernel Hierarchical Extreme Learning Machine (KH-ELM). The dataset used for this purpose comprises rs-fMRI scans of 68 subjects, including 23 ADHD patients and 45 TDCs, and FC features are extracted with a hierarchical ELM sparse autoencoder. For classification 5-fold CV is employed, resulting in an impressive accuracy of 97.17%. Zhao et al. [69] introduced a novel dynamic graph convolutional network (dgCN) to

improve ADHD diagnosis. The methodology involves utilizing rs-fMRI scans obtained from the ADHD-200 database, which comprises scans of 635 subjects including 260 ADHD patients and 343 TDCs with age ranges between 7 to 20 years. Subsequently, FC features are extracted from the brain functional graph, and classification is performed using the dgCN model with 10-fold CV. The experimental results reveal that this model outperforms others, achieving an accuracy of 72%.

Additionally, Qiang et al. [70] proposed the Spatiotemporal Attention Auto Encoder (STAAE) for the classification of ADHD using Resting State Temporal Templates (RSTT). For experimentation, they employed rs-fMRI data from the ADHD-200 dataset, comprising 620 subjects which have 260 ADHD patients and 340 TDCs, age ranges between 7 to 21 years. Temporal features are then extracted from data and classification is conducted using STAAE with 5-fold CV, resulting in an accuracy of 72.5%. Meanwhile, Qin et al. [71] proposed a technique for diagnosing ADHD by integrating rs-fMRI data and its relevant phenotypic\PC data through stacking, an ensemble learning approach. They collected these datasets from the ADHD-200 database, containing data from 797 subjects, which have 375 ADHD patients and 422 TDCs, aged between 7 to 26 years. Subsequently, spatio-temporal features are extracted from rs-fMRIs utilizing Trans-3D while clinical features are extracted from PC data employing RF. Classification is carried out using an ensemble learning model, achieving an accuracy of 74.5%. Furthermore, Chen et al. [72] presented an Attention Autoencoding Neural Network (Att-AENet) for detecting FC biomarkers for ADHD diagnosis. They employed rs-fMRI scans from the ADHD-200 database which contains data of 578 subjects, including 242 ADHD patients and 336 TDCs aged between 7 to 18 collected. FC features are extracted and detection is carried out using ATT-AENet. The experiential results utilizing LOOCV indicate that the model outperforms others, achieving 98.9% accuracy.

Peng et al. [73] proposed a Summation-based Synergic Artificial Neural Network (SSANN) for diagnosing ADHD. This approach employed dual 3DCNN branches to extract features from sMRI scans and rs-fMRI scans. These scans are obtained from the ADHD-200 dataset, which includes information from 840 subjects, consisting of 310 ADHD patients and 520 TDCs with age ranges between 7 to 21 years. Subsequently, these features are integrated with the proposed model and fed into the fully connected NN for classification. The data is split into an 80:20 ratio for training and testing, resulting in an accuracy of 72.89%. Moreover, Anitha and Geroje [74] explore machine learning approaches such as SoftMax Regression and SVM for the identification and classification of ADHD, by utilizing rs-fMRI data. This data sourced from the ADHD-200 dataset, contain information from 164 subjects aged between 7 to 22 years. Subsequently, spatio-temporal features are learned from raw data and classification is conducted using the hold-out method, resulting in an impressive accuracy of 94%. Singh et al. [75] utilize NB for the identification of ADHD by

TABLE 6. Summary of studies that utilize MRI scans for the diagnosis of ADHD.

Sr. #	Ref.	Type of MRI Scans	Dataset	Age	Features	ML/DL Model	Validation Approach	Accuracy %
1	[66]	rs-fMRI	20 ADHD, 20 TDCs	N/A	RSN	BLSTM	80:20	87.50
2	[67]	rs-fMRI	600 ADHD, 600 TDCs	7 to 26	MRI Images	CNN	70:30	97.78
3	[68]	rs-fMRI + PC	400 overall	7 to18	PC + FC	Ensemble (NN & SVM)	70:30	99
4	[69]	rs-fMRI	328 overall	7 to 21	Temporal	3D CNN, 3DGAN	55:45	97
5	[70]	rs-fMRI	23 ADHD, 45 TDCs		FC	KH-ELM	5-fold CV	97.17
6	[71]	rs-fMRI	260 ADHD, 343 TDCs	7 to20	FC	dGCN	10-fold CV	72
7	[72]	rs-fMRI	260 ADHD, 340 TDCs	7 to21	Temporal	STAAE	5-fold CV	72.5
8	[73]	rs-fMRI + PC	375 ADHD, 422 TDCs	7 to26	MRI Images, + clinical	stacking (Trans3D, RF), MLP	CV	74.5
9	[74]	rs-fMRI	242 ADHD, 336 TDCs	7 to18	FC	Att-AENet	LOOCV	98.9
10	[75]	rs-fMRI+sMRI	310 ADHD, 520 TDCs	7 to26	MRI Images	SSANN	80:20	72.89
11	[76]	rs-fMRI	164 overall	7 to 22	MRI Images	SoftMax Regression, SVM	holdout	94
12	[77]	PC	N/A	N/A	PC	NB	N/A	82.5
13	[78]	rs-fMRI	351 ADHD, 430 TDCs	7 to21	MRI Images	2D CNN, 2DCNN-LSTM		98.12
14	[79]	rs-fMRI + PC	247 ADHD, 338 TDCs	7 to18	PC + FC	auto-encoding network	LOOCV	99.7
15	[80]	rs-fMRI	352 ADHD, 429 TDCs	7 to22	ROI based	SCCNN-RNN	leave one site out	70.46%
16	[81]	rs-fMRI	20 ADHD 20 TDCs	8 to 21	MRI Images	CNN	N/A	86%
17	[82]	MRI	1359 Overall	N/A	MRI Images	BoostAlexNet	5-fold CV	93.67
18	[83]	rs-fMRI	295 ADHD, 364 TDCs	7 to 21	FC	NN Autoencoder, SVM, DT, LD, RF	10-fold CV	73.3
19	[25]	s-MRI + PC	316 ADHD, 316 TDCs	7 to 22	PC + ROI Based	KNN, LR, RF, SVM, RBSVM, mRMR, EFS	10-fold CV	75%
20	[26]	rs-fMRI	99 ADHD, 150 TDCs	8 to17	MRI images + ROI based	CNN, autoencoder	Holdout	69%
21	[84]	rs-fMRI	357 ADHD, 523 TDCs	7 to 26	FC	CNN, GAT	N/A	70.73
22	[85]	rs-fMRI	357 ADHD, 582 TDCs	7 to 26	FC	Transformer with defision kernel	80:20	78.4
23	[86]	rs-fMRI	118ADHD, 98 TDCs	8 to15	FC	Multi keranl SVM	5-fold CV	72.3
24	[87]	rs-fMRI + PC	319 ADHD, 528 TDCs	9 to16	PC + FC	CNN with hash layer	5-fold CV	69.09
25	[88]	rs-fMRI, PC	358 ADHD, 581 TDCs	7 to 26	ROI based	LSTM , transfer learning	10-fold CV	72.62
26	[89]	rs-fMRI	822 overall 38 overall	7 to 24 10 to11	Unstable DMs/DMs, eigenvalues, eigenvectors	MLP, SVM, KNN, NB, LR, RF, SGD, BN, AdaBoost	10-fold CV	94.7, 73.2
27	[90]	rs-fMRI	(112 ADHD, 77 TDC)Adults (106 ADHD73 TDCs) children	22 to30 8 to10	FC	XGBoost, GRadBoost, AdaBoost, Extra Trees, RF, FS-RIWELL	Nested 3-fold CV	80
28	[91]	rs-fMRIs	ADHD 170, TDCs 182	7 to21	FC	1D NN, 2D NN	N/A	90.90
29	[92]	PC	285ADHD, 491 TDCs	7 to 21	Clinical	LR, SVM, DT, RF, KNN, MLP	N/A	N/A
30	[93]	rs-fMRI	353 ADHD, 489 TDCs	7 to 21	FC	ASTNet	80:20	90.9
31	[94]	rs-fMRI	231ADHD, 232 TDCs	8 to 22	faLFF, ReHo	3D-CNN	5-fold CV	76.19

employing preprocessed extracts from rs-fMRI data sourced from the ADHD-200 dataset. This data contains 9 PC features and the experimental result reveals that this model outperforms others, achieving 82.5% accuracy. Meanwhile, Khullar et al. [76] aimed to classify ADHD patients from

TDCs using deep learning models such as 2D CNN and a hybrid 2DCNN-LSTM. The methodology involves employing rs-fMRI data from the ADHD-200 dataset consisting of 781 subjects, comprising 351 individuals with ADHD and 430 TDCs aged between 7 to 21 years. Subsequently,

spatio-temporal features are extracted from images and classification is performed using DL models, resulting in an impressive accuracy of 98.12%. Tang et al. [77] proposed a modified auto-encoding (AE) network for the classification of ADHD using rs-fMRI data along with relevant PC information. They sourced this data from the ADHD-200 dataset, containing a total of 585 subjects, consisting of 247 ADHD individuals and 338 TDCs aged between 7 to 18 years. Extracted features include FC and PC, and classification is conducted using proposed model with LOOCV. The results show that this model performs excellently, achieving an accuracy of 99.7%. Kim et al. [78] used deep learning models for the classification of ADHD by focusing on essential brain regions. They collected rs-fMRI data from the ADHD-200 dataset, which contains 781 subjects' data, comprising 352 ADHD patients and 429 TDCs with age ranges between 7 to 21 years. The Region of Interest (ROI) based spatio-temporal features are extracted from 15 regions of the brain and 'Separate Channel' CNN-RNN (SCCNN-RNN) is implemented as the base model architecture. Training and testing are performed using LOOCV, resulting in an accuracy of 70.6%. Additionally, De Silva et al. [79] conducted a study focusing on the identification of ADHD using rs-fMRI data, by evaluating various feature extraction and selection methods such as SBC, fALFF, and ReHo. The data is obtained from the ADHD-200 dataset, which includes 40 participants, with 20 in each category (ADHD and TDCs) between the ages of 8 to 21 years. The spatio-temporal features are extracted from images and classification is performed using CNN. The results indicate that the best accuracy, which is 86% achieved with SBC. Sheriff and Gayathri [80] presented an ensemble approach BoostAlexNet comprising various trained models such as ResNet 101, NASNet, Xception, MobileNet, and Inception V3 for the classification of ADHD. They collected MRI data from the Kaggle repository, which consists of 1359 images, subsequently extracted features from these images, and performed classification using BoostAlexNet. The training and testing are carried out with a 5-fold CV, resulting in an impressive accuracy of 93.67%.

Furthermore, Khan et al. [81] proposed a knowledge distillation approach to find discriminative features between TDCs and ADHD. The methodology involves training a large NN on FC features extracted from rs-fMRI data, containing information from 295 individuals with ADHD and 364 TDCs aged between 7 to 21 years. Subsequently, an autoencoder is employed to reproduce embedding, feedforward approach is utilized for feature selection, and knowledge distillation is implemented. The training and testing are conducted using 10-fold CV and various models such as SVM, RF, DT, and LD are applied to evaluate the efficiency of the feature selection approach. Results indicate the maximum achieved accuracy is 73.3%. Lohani et al. [24] introduced an automated approach for the diagnosis of ADHD using sMRI data along with relevant PC data. For this purpose, they obtained data from the ADHD-200 dataset, consisting of 632 participant's data, with 316 belonging to each category (ADHD and TDCs)

aged between 7 to 22 years. ROIs-based and PC features are extracted followed by feature selection using Minimum Redundancy Maximum Relevance (mRMR) and EFS. Subsequently, classification is carried out utilizing five popular classifiers, such as linear SVM, KNN, LR, RBSVM, and RF. A 10-fold CV approach is employed for training and testing and the maximum achieved accuracy is 75% with RBSVM and SVM. Wang et al. [25] presented two novel deep learning-based methods: Independent Component Analysis (ICA) with CNN and the correlation Autoencoder for ADHD classification. They employed rs-fMRI data from the ADHD-200 dataset, which consists of 219 subject's data, comprising 150 TDCs and 99 ADHD patients with age ranges between 8 to 17 years. Features from overall images and ROIs are extracted and classification is performed by proposed methods using a hold-out validation approach. Results reveal the maximum achieved accuracy is 69%. Meanwhile, Pei et al. [82] conducted a study in which they introduced three data augmentation methods: Gaussian noise, Mix-up, and Sliding Window alongside a deep feature fusion method by integrating CNN and GAT to improve the identification performance of the disease. The data utilized here is rs-fMRI collected from ADHD-200, containing information from 880 participants, which include 523 TDCs and 357 ADHD patients aged between 7 to 26 years. FC features are extracted from the data and 70.73% accuracy is achieved with the sliding window method. Zhang et al. [83] present the use of a transformer for combining function brain network modeling, analysis, and classification of brain disorders employing the Diffusion Kernel Attention network. The dataset utilized in this study is rs-fMRI obtained from the ADHD-200 dataset, encompassing information from 939 subjects, which contain 357 ADHD individuals and 582 TDCs with age ranges between 7 to 26 years. FC features are extracted and an 80:20 ratio split is used for training and testing. The experimental results indicate that the proposed method achieves 78.4% accuracy. Similarly, Wang et al. [84] proposed a Distribution Guided Network Thresholding method for examining functional connectivity based on rs-fMRI data for brain disorder classification. They sourced data from the ADHD-200 dataset, which consists of 216 participants' data, including 118 ADHD patients and 98 TDCs aged between 8 to 15 years. FC features are extracted and classification is carried out using multi-kernel SVM with 5-fold CV, resulting in 72.3% accuracy. Ji and Zhang [85] based on deep graph hashing learning, introduced a novel brain network classification method known as BNC-DGHL. The methodology involves employing rs-fMRI data along with relevant phenotypic\PC data from the ADHD-200 database. This dataset consists of 847 subjects' data, including 319 ADHD patients and 528 TDCs aged between 7 to 26 years. Deep features (FC and phenotypic\PC) are extracted and classification is performed using CNN with a hash layer. 5-fold CV is carried out for training and testing, resulting in an accuracy of 69.09%. Hu et al. [86] proposed Source Free Semi-Supervised Transfer Learning (S3TL)

for the identification of mental disorders. The methodology involves utilizing rs-fMRI data from the ADHD-200 dataset, encompassing data of 939 participants, including 358 ADHD patients and 581 TDCs aged between 7 to 26 years. ROI-based spatio-temporal features are extracted and classification is carried out by the proposed method which utilizes LSTM and transfer learning. 10-fold CV is performed, resulting in an accuracy of 72.62%.

Moreover, Ingabrie et al. [87] analyze the stability of decomposed fMRI BOLD signals for the detection of disease. This study utilized rs-fMRI data collected from the ADHD-200 dataset composed of 882 subjects aged between 7 to 24 years and from the Huaxi Hospital dataset containing data from 38 participants aged between 10 to 11 years. Three types of features are extracted: from the ratio of unstable dynamic mode (DMs) to the complete set of DMs, from eigenvalues, and eigenvectors. Subsequently, nine well-known classifiers (Bayesian Network (BN) MLP, SVM, KNN, NB, LR, RF, SGD, AdaBoost) with 10-fold CV are employed to classify TDCs and ADHD patients. The results show that the maximum accuracy achieved on both datasets is 94.7% and 73.2% respectively. Furthermore, Samar [88] developed an integrated model for the classification of ADHD by utilizing an integrated feature selection framework. For this purpose, they employed rs-fMRI data from the Hospital of Peking University (PKU part of ADHD-200) and Beijing Normal University (BNU). It contains data from 189 adults and 179 children participants with age ranges between 20 to 30 and 8 to 10 years respectively. FC features are extracted, and subsequently, selected by the novel framework FS-RIWELL. Ensemble classifier with XGBoost, GRadBoost, AdaBoost, Extra Trees, and Random Forest employed for classification. Training and testing are carried out with nested 3-fold CV, resulting in an accuracy of 80%. Agarwal et al. [89] conducted a study, in which they investigated the accuracy of ADHD detection on rs-fMRI data employing deep learning. The data is sourced from the ADHD-200 dataset, comprising 352 subjects, 170 ADHD patients, and 182 TDCs aged between 7 to 21 years. FC features are extracted and two deep learning models, 1D NN, and 2D NN are employed for classification. Experimental results indicate that 90.90% accuracy is achieved. Moreover, Singh et al. [75] investigate various machine learning techniques for the diagnosis of ADHD by extracting personal characteristics (PC) data from the ADHD-200 dataset. This dataset contains data from 776 participants, 285 ADHD patients, and 491 TDCs aged between 7 to 21 years. PC features are extracted and then selected by using Cfs-SubsetEval, GA, and WrapperSubsetEval techniques. For classification six popular classifiers (LR, SVM, DT, RF, KNN, MLP) with 10-fold CV are utilized, resulting in an accuracy of 77.84%. Qiu et al. [90] introduced a deep learning-based approach namely Adaptive Spatial-Temporal Neural Network (ASTNet) for diagnosing ADHD using rs-fMRI data. The data collected from the ADHD-200 database includes 353 ADHD patients and 489 TDCs with ages

TABLE 7. Summary of the ADHD-200 dataset.

Sr.#	Imaging site	ADHD Patients	TDCs
1	Bradley Hospital/Brown University	24	27
2	Kennedy Krieger Institute	22	61
3	NeuroIMAGE sample	25	23
4	New York University Child Study Center	123	99
5	Oregon Health Sciences University	37	42
6	Peking University	113	146
7	University of Pittsburgh	6	95
8	Washington University at Saint Louis	0	61
9	Total	350	554

ranging from 7 to 21 years. This model adapts dynamic Functional Connectivity (dFC) and captures global temporal dependencies. Moreover, provides insight into function activation correlation and automated ADHD classification with an 80:20 train-test split ratio. The experimental results indicate that the proposed model achieves 90.9% accuracy. Gulhan and Özmen [91] proposed an automated ADHD detection approach using 3D-CNN based on fMRI regional analysis. The data sourced from the ADHD-200 dataset, encompassing 231 ADHD patients and 233 TDCs with ages ranging from 8 to 22 years. They selected the fractional amplitude of low-frequency fluctuations (faLFF) and regional homogeneity (ReHo) data and performed classification with 3D-CNN with a five-fold CV. The approach achieves a maximum accuracy of 76.19%.

From TABLE 6 overall analysis of these MRI-based studies indicates that 31 out of 98 studies used this modality for the diagnosis of ADHD. Moreover, DL approaches occupied a bigger portion as compared to ML, where 21 out of 31 studies utilized DL techniques, while 5 employed ML, and 5 incorporated both DL and ML approaches. Additionally, to obtain data most of the studies utilized a public database: ADHD-200,³ provided by Neuro Bureau, as out of 31 studies, 28 utilized this dataset. However, almost each study selected data from varying number of participants and varying number of imaging sites. ADHD-200 is a consortium that collected structural MRI scans and resting-state functional MRI scans along with relevant PC data from 554 TDCs and 350 ADHD children/ adolescents. Overall, 8 websites are involved in the collection of this data and a summary is presented in TABLE 7. Furthermore, brain Functional Connectivity (FC) features emerged as the most common features for the identification of ADHD, as 13 out of 31 studies employed these features as input to the AI models.

C. QUESTIONNAIRE/ RATING SCALES

In the current research, the medical researchers still employed questionnaires or rating scales as a diagnostic tool for

³http://fcon_1000.projects.nitrc.org/indi/adhd200/

the diagnosis of ADHD. In TABLE 8, the investigation of ADHD diagnosis based on questionnaires or rating scales is performed according to the parameters presented below. **Questionnaires** indicate the types of rating scales or questionnaires employed as a diagnostic tool. We try to cover those questionnaires that studies utilized to develop their best-performing models, but in most of the studies, all types of questionnaires are deemed equally important, hence we include them all in our analysis. Next, **Dataset** indicates the size of the data employed, and **Age**, refers to the age of participants from whom data has been collected. Moreover, **ML/DL Model** shows the type of model used for ADHD detection or classification, whether it's ML, DL, or both. Furthermore, the **Validation Approach** indicates the train test split used for classification and lastly, **Accuracy %** refers to the maximum accuracy obtained by any approach in percentage. The summary of these research articles is provided in the following paragraphs.

In TABLE 8, Goh et al. [92] aimed to develop a data-driven method based on machine learning for ADHD detection. For this purpose, they collected data from 399 participants, including 222 with ADHD and 177 not meeting the diagnostic criteria. The age of these participants is 7 to 13 years in Year 1, and 11 to 18 years in Year 6. ADHD symptoms are evaluated by using the KSADS for parent reporting and ADHD-RS for teacher reporting. After evaluating the data based on diagnostic criteria, it has been utilized as input to the Random Forest for classification with a 5-fold CV, resulting in a maximum of 93% accuracy. Chen et al. [93] proposed a hybrid Artificial Intelligence (AI) model for the detection of ADHD. The dataset utilized in this study consists of 501 anonymized assessments for ADHD patients, including 265 TDCs and 236 ADHD patients, aged between 17 to 72 years, and obtained from 2019 to 2022 July. This dataset is collected by using some screening tools and evaluated by a diagnostic tool namely, Diagnostic Interview for ADHD in adults (DIVA). Afterward, passed as input to a knowledge-based model, Decision Tree model, and a hybrid model combining both approaches, resulting in a maximum accuracy of 93.61%. Moreover, Grazioli et al. [94] explored the usability of telemedicine for the diagnosis of ADHD and ASD based on supervised machine learning. They collected the data from 342 subjects with age ranges between 3 to 16 years. For the evaluation of this data various diagnostic measures have been utilized, including, case history and demographic questionnaire, Conner's Parent Rating Scale-Revised (CPRS-R), Child Behavior Checklist (CBCL), Social Responsiveness Scale (SRS) and Conner's Teacher Rating Scale-Revised (CTRS-R). Subsequently, the extracted features from the data are passed to the Decision Tree, a classifier employed for classification with LOOCV. The results indicate that the proposed method achieves 74% accuracy.

Liu et al. [95] conducted a study to develop a screening tool for ADHD and comorbid mental health disorders based on machine learning. They obtained data from 955 adults including 432 with ADHD and 532 without this disorder with

an average age of 35.35 years. The participants completed the Early Detect Questionnaires (a set of self-report clinical questionnaires) and then checked by certified psychiatrists of the center. Afterward, the data is passed as input to the classifier Elastic Net for classification using 5-fold CV and LOOCV. The results show that the proposed method achieved an accuracy of 78.8%. Similarly, Chatpreecha and Usanavasin [96] also aimed to design and develop a screening tool for teachers, parents, and doctors to observe ADHD symptoms in children and classify ADHD using machine learning. The data gathered from questionnaires completed by teachers and parents utilizing the Vanderbilt ADHD Diagnostic Rating Scale (VADRS), this data consists of 420 children with ages ranging between 6 to 12 years. Subsequently, after receiving the data it is fed to classifiers such as DT, NB, NN, and KNN for classification using an 80:20 split ration. The results reveal that the proposed tool achieves a very high 99.60% accuracy.

Additionally, Jafari et al. [97] analyzed the differential item functioning (DIF) in the child and parent reports of the KINDL measure to diagnose ADHD. The data sample consists of 122 children with ADHD and 1086 TDCs, aged between 8 to 16 years meanwhile, 127 parents with ADHD and 1061 TDCs respectively. The data is evaluated by using the KINDL quality of life questionnaire and passed as input to the classifier named Generalized Partial Credit Model with LASSO penalization (GPCMlasso) for classification with a 5-fold CV. Maniruzzaman et al. [98] conducted a study to find risk factors for children with ADHD using machine learning. The data is extracted from the National Survey of Children's Health (NSCH) spanning from 2018 to 2019, comprised of 45,779 children, among which 5218 with ADHD and the rest of TDCs with age ranged between 3 to 17 years. Afterward, this data inputted to classifiers such as RF, SVM, KNN, DT, NB, MLP, 1DCNN, and XGBoost for classification using 10-fold CV, achieving 85.5% of accuracy. Similarly, machine learning is also utilized by Chen et al. [99] developed a diagnostic model for detecting ADHD. The methodology involves obtaining data from A National Health Service specialist provider, between 2014 to 2017 which contained 69 such patients with an average age 9.931 years. The screening questionnaires encompass Conner's ADHD Rating Scales, the Iowa Personality Disorder Screen (IOWA), the Drug Abuse Screening Test (DAST-10), the Alcohol Use Disorder Identification Test (AUDIT), Mode Disorder Questionnaire (MDQ), the GAD-7 measuring GA and the Patient Health Questionnaire (PHQ-9). Moreover, the clinical interviews are both structured such as DIVA, and unstructured such as the text of the final medical report. Subsequently, an assessment-centered dataset is passed to six well-known classifiers, including SVM, LR, NB, RF, DT, and KNN with LOOCV, resulting in an accuracy of 85.507%. Moreover, Kim et al. [100] investigated the use of machine learning with Minnesota Multiphasic Personality Inventory-2 (MMPI-2) to predict symptoms of ADHD in adults. The data is obtained from college students, consisting of 5726 participants with an average age of 19.8 years. The MMPI-2-Restructured Form is

TABLE 8. Summary of studies that utilize questionnaire/ rating scale for the diagnosis of ADHD.

Sr. #	Ref.	Questionnaires	Dataset	Age	ML/DL Model	Validation Approach	Accuracy %
1	[95]	KSADS, ADHD-RS	222 ADHD, 177 TDCs	7 to 13, 11 to 18	Random Forest	5-fold CV	81-93
2	[96]	DIVA	236 ADHD, 265 TDCs	17 to 72	DT	10-fold CV	75.03
3	[97]	CHDQ, CPRS-R, CBCL, SRS, CTRS-R	342 overall	3 to 16	DT	LOOCV	74
4	[98]	Early Detect Questionnaires	432 ADHD, 523 Non-ADHD	35.35	Elastic Net	5-fold CV, LOOCV	78.8
5	[99]	VADRS	420 Overall	6 to 12	DT, NB, NN, KNN	80:20	99.60
6	[100]	KINDL	122 ADHD, 1086 TDCs, 127 ADHD 1061 TDCs	8 to 16	GPCMlasso (Statistical)	5-fold CV	N/A
7	[101]	NSCH	40561 TDCs, 5218 ADHD	3 to 17	RF, SVM, KNN, DT, NB, MLP, 1DCNN, XGBoot	10-fold CV	85.5
8	[102]	DAST-10, IOWA, AUDIT, MDQ, GAD-7, PHQ-9	69 overall	9.931	SVM, LR, NB, RF, DT, KNN	LOOCV	85.507
9	[103]	ASRS, MMPI-2-RF	5726 overall	19.8	KNN, LDA, RF	80:20	93.6
10	[104]	WISC-R, SDF, FAF, Temps-A, BIS/BASS	60 ADHD, 60 TDCs	8 to 12	DT, LSTM, SVM, LR, 1DCNN	5-fold CV	92.5
11	[105]	CDIS	221 overall	N/A	RF, DT, MLP, SVM	N/A	91

utilized and symptoms of ADHD are evaluated by employing the ADHD Self-Report Scale (ASRS). Subsequently, classification is performed using KNN, LDA, and RF with 58% for training and 20% for testing, resulting in an accuracy of 93.6%. Furthermore, Altun et al. [101] examined the applications of deep learning and classical machine learning based on temperament features. The data utilized in this study comprises 120 participants with 60 in each category: ADD and TDCs, aged between 8-12 years. Additionally, various evaluation criteria are utilized to assess the data including the sociodemographic data form, Conners family assessment form, Temperament scale for school-age children, Temps-A temperament scale, Behavioral inhibition system/ behavioral activation system scale, and WISC-R intelligence test. Afterward, temperament features are extracted and selected using PCA. Classification is carried out using DT, LSTM, SVM, LR, and 1DCNN with a 5-fold CV. Results indicate that this study achieved a maximum of 92.5% accuracy with DT. Lastly, machine learning is also employed by Alsharif et al. [102] to diagnose ADHD. The study collected ERP data from 221 participants and evaluated using the Clinical Diagnostic Interviewing Scale (CDIS). To address the issue of data imbalance, two techniques have been applied: Oversampling and Under-sampling. Classification is performed using RF, DT, MLP, and SVM, resulting in an accuracy of 91%.

From TABLE 8, the overall investigation of these questionnaire-based studies shows that only 11 out of 98 selected studies employed this diagnostic method for the diagnosis of ADHD. In this section, the majority of studies utilized ML models, with 6 out of 11 employing these models, 4 studies utilized both ML and DL models, 1 study used a statistical model while none solely relied on DL models. It can also be seen that DT is the most commonly employed model in 7 out of 11 studies and RF is the second most utilized

model appearing in 5 out of 11 studies for the classification of questionnaire-based data. Furthermore, in TABLE 8, no questionnaire type can be identified as predominantly utilized as a diagnostic tool. Each questionnaire appears once or twice, indicating that all are equally important. Researchers, likely select them based on their specific needs or the attributes of the data they are investigating.

D. MOTION DATA/ACTIVITY RECORDING

Research studies have employed motion data or data recorded during different activities as a diagnostic tool for ADHD diagnosis. For this purpose, various recording devices are utilized, further information is provided in TABLE 9, based on some parameters. Initially, the **Recording Device** indicates the device that is normally worn by participants to record the data. Subsequently, the **Dataset** represents the size of recorded data while **Age** represents the age range of participants. **Features** indicate the extracted features from the data fed to the classification models and the **ML/DL model** refers to the classifiers utilized in these studies for ADHD classification. Furthermore, **Validation Method** shows the train-test split ratio. Lastly, **Accuracy %** indicates the maximum accuracy achieved by any model, represented as a percentage. The summary of these studies is provided in subsequent paragraphs.

In TABLE 9, Amado-Caballero et al. [103] introduced an approach for ADHD diagnosis with deep learning on actimetry. The actimetry recordings are obtained with an ActiGraph GT3x device placed in the non-dominant wrist. These recordings collected from 139 subjects, comprising 70 ADHD patients and 69 TDCs with age ranges between 6 to 15 years. All subjects monitored in their daily activity for approximately 24 hours. From this data 2D time-frequency images are created and used as input for feature extraction.

TABLE 9. Summary of studies that utilize motion data or activity recording for the diagnosis of ADHD.

Sr.#	Ref.	Recording Device	Dataset	Age	Features	ML/DL Model	Validation Method	Accuracy %
1	[106]	ActiGraph GT3x Device	70 ADHD, 69 TDCs	6 to 15	2D Time-frequency Image	CNN, SVM	N/A	N/A
2	[107]	ActiGraph GT3x Device	39 overall	7 to 16	Demographic, Clinical, Sensor-Derived	Random Forest	70:30	82
3	[108]	N/A	12348 ADHD, 39160 sleep problems	9 to 11	circadian rhythm-based features, demographic	RF, extreme GB, light GB	70:10:20	AUC. 0.798
4	[109]	Actigraphy Device,	51 ADHD, 52 Clinical Controls	N/A	activity-related statistical features	C4.5, KNN, RF, Logit Boost, SVM, NB, PCA	10-fold CV	98.43
5	[110]	Actigraphy Device,	45 ADHD, 71 TDCs	N/A	activity-related statistical, PC	SVM, RF, KNN, DT, MLP, CfsSubsetEval	LOOCV	98
6	[111]	Actigraphy Device,	114 samples	N/A	N/A	Several Models	10-fold CV	99.2
7	[112]	LemurDx App	15 ADHD, 15 TDCs	6 to 11	Time-based	RF, RLR	LOOCV	89
8	[113]	head-mounted device	1,000 data points	N/A	N/A	CNN	70:30	98.3

Subsequently, classification is carried out using CNN and SVM, and Occlusion maps are employed to highlight patterns explaining ADHD activity. Park et al. [104] proposed a machine learning-based ADHD detection approach using sensor-based activity monitoring. They obtained recordings from 39 participants, aged between 7 to 16 years. Participants need to wear a waist-worn ActiGraph GT3x device for up to one week, three times over 12 months. Afterward, demographic, clinical, and sensor-derived features are extracted and utilized as input to a random forest classifier for classification. A 70:30 split ratio is employed for training and testing, resulting in an accuracy of 82%.

Moreover, Kim et al. [105] introduced a machine learning-based approach for detecting ADHD and sleep problems using wearable data. The dataset utilized in this study is obtained from an ABCD study comprising screening data from 6571 patients and 21 days of wearable data from 5725 patients, aged between 9 to 11 years. A total of 12348 wearable data for ADHD and 39160 for sleep problems is utilized, and demographic and circadian rhythm-based features are extracted. Subsequently, a 70:10:20 split ratio is employed for training, validation, and testing, and classification is carried out with three well-known classifiers such as RF, extreme GB, and light GB, resulting in AUC 0.798.

Kaur and Kahlon [106] presented a study aimed at the accurate identification of ADHD among adults using real-time activity data. They utilized the publicly available Hyperactiv dataset, comprising health, activity, and heart rate data from 103 patients, including 51 ADHD patients and 52 clinical controls. This dataset contains data recordings of motor activity on human rest and activity cycle obtained using a wrist-worn actigraphy device, and the heart rate variability using a chest-worn ECG device. Activity-related statistical features have been extracted followed by feature selection using PCA. Subsequently, the selected features used as input to six popular classifiers C4.5, KNN, RF, LogitBoost,

SVM, and NB for classification with 10-fold CV, achieving an accuracy of 98.43%. Similarly, Mohd et al. [107] also introduced a machine-learning approach for detecting ADHD using the Hyperactiv dataset. The data consists of 116 patients, encompasses 45 ADHD patients and 71 TDCs. Activity-based statistical and PC features are extracted from the data. Afterwards, features are selected using CfsSubsetEval and classification is carried out using SVM, RF, KNN, DT, and MLP with LOOCV. The results indicate that the proposed approach achieves 98% accuracy. Deshmukh et al. [108] explore brain regions' contributions to ML-based classification of ADHD using EEG signals. They sourced the signals from the IEEE data port, extracted statistical features, and performed classification using six well-known classifiers: NB, RF, DT, KNN, SVM, Adaboost, and LDA with a 5-fold CV. The maximum achieved accuracy is 84% with RF. Furthermore, Lindhiem et al. [109] explored the diagnosis of ADHD using machine learning and mobile sensing. The data utilized in this study is recorded by wearing an Apple watch with the LemurDx app, and it is worn by 28 out of 30 participants, along with contextual data and activity labels. After data collection, time-based features are extracted and Randomized Logistic Regression (RLR) is utilized to select features. Subsequently, classification is performed using random forest with LOOCV, resulting in an accuracy of 89%.

Son et al. [110] proposed a novel approach based on Virtual Reality (VR) and machine learning for ADHD detection. The dataset employed in this study comprises sensor data recorded by using a head-mounted device with speakers attached near the ears from VR scenes and potentially from experiments involving both ADHD and non-ADHD participants. Subsequently, 100 data points are created from this data, each treated as a potential patient and fed into a CNN for classification with a 70-30 split. The results indicate that this novel approach performs excellently with 98.3% accuracy.

From TABLE 9, the overall examination of these activity recording researches indicates that only 8 out of 98 selected

studies utilized this diagnostic tool for the identification of ADHD. Subsequently, the ML models applied in 4 studies, ML and DL both employed in 3 studies while DL utilized only in 1 study. It can also be evident from TABLE 9, that RF is the most utilized model, appearing in 5 out of 8 studies for the classification of activity recording-based data. In terms of activity recording devices, Actigraphy devices emerge as the most utilized with 5 out of 8 researchers employing them for activity recording. Specifically, the ActiGraph GT3x is in common use, this lightweight, compact user-friendly device with a battery life of up to 25 days [105]. Furthermore, in terms of datasets, 2 studies utilized activity recordings obtained from a publicly available database known as Hyperaktiv,⁴ which encompasses health, activity, and heart rate data.

E. OTHERS

Researchers also used other modalities for the identification of ADHD which are less common but still in use. These modalities include gaming data, pupillometric data, eye-tracking, etc. In TABLE 10, the analysis of ADHD diagnosis based on some least employed modalities utilized in one, two, or a maximum of three selected studies is conducted, according to the parameters presented below. Firstly, **Modality** represents the type of diagnostic method used for the diagnosis of ADHD. **Dataset** shows the size of data consists of ADHD patients and TDCs or patients of some other neurodevelopmental disorder while, **Age** indicates the age of participants, who participated in the data collection process. Moreover, **Features** represents the characteristics extracted from data and used as input to the ML\DL model. **ML\DL Model** indicates either ML, DL, or both types of classifiers used for the classification\ detection of ADHD. **The validation Approach** refers to the train-test split ratio and lastly, **Accuracy %** represents the highest accuracy achieved by any model in percentage. The summary of these studies is provided in subsequent paragraphs.

In TABLE 10, Lee et al. [111] presented a novel approach for ADHD screening and diagnosis in children by using skeleton data obtained through a game. This data is collected from 596 children from 8 to 13 years old, among these children 66 are ADHD patients, 181 are at risk and 349 are TDCs. Subsequently, a total of eleven data types are used and fed to three well-known classifiers such as GRU, RNN, and LSTM with a bidirectional layer for classification using LOOCV. The experimental results reveal that LSTM outperforms as compared to others with 97.82% accuracy. In another study [112] they utilized the same data and successfully classified TDCs, ADHD, and ADHD-Risk groups with a higher accuracy of 98.15% providing valuable insight to teachers and clinicians. Jaiswal et al. [113] proposed a methodology that involves in the development of a gamified web system for data collection, fusion of crowdsourcing algorithms with ML behavioral feature extraction, development of DL models for the

classification of ADHD and ASD, and implementation of an active learning system to query for additional data. There are a total of 400 participants recruited in this study, which encompasses 100 ADHD patients, 100 ASD patients, 100 ADHD+ASD, and 100 TDCs with age ranges between 14 to 18 years.

The video and audio data obtained through this gamified web system feed to the LSTM for behavioral features extraction and classification using 60:20:20 train, validation, and test ratio. The results show that the proposed method achieves 71% accuracy. Vimalajeewa et al. [114] introduced a wavelet approach for diagnosing ADHD using pupil diameter time series data. The dataset obtained from a public repository featured in Rojas-Libano et al. collected from 50 children, comprising 28 ADHD patients and 22 TDCs aged 10 to 12 years. Wavelet domain spectral features are extracted from this data using Fisher criteria and feed to the three popular classifiers: LR, SVM and KNN for classification with 67:33 training and testing split ratio. The results show that this method achieves 84.3% of accuracy. Das and Khanna [115] presented another approach for ADHD detection using same pupil diameter time series data. After obtaining the data a total of 783 features are engineered including absolute energy, consecutive quintile changes, Fourier transform metrics, approximate entropy, pupil size dilation velocity as well as aggregated linear trends, out of which 218 found to be statistically significant. Subsequently, these features are fed into six popular classifiers: LR, KNN, NB, DT, RF, and SVM for classification with a 10-fold CV. The results indicate that SVM performs excellent with an accuracy of 85.6%.

Yoo et al. [116] developed an approach for ADHD screening using eye-tracking data. The data is collected from 135 participants using digital devices during the performance of five behavioral tasks, these participants include 56 ADHD patients and 79 TDCs aged 6 to 12 years. Subsequently, eye tracking features are extracted and passed to two classifiers: RF and ExtremeTrees for classification with a 5-fold CV resulting in an accuracy of 76.3%. Shin et al. [117] utilized handwriting patterns and machine learning models for the diagnosis of ADHD and ASD. They collected the handwriting pattern samples from 29 children (14 having ADHD with coexisting ASD and 15 are TDCs) using pen tablets. The age of patients ranges from 5 to 14 and TDCs are 8 to 12 years. Afterward, they extracted statistical features from these patterns and passed them to Sequential Floating Forward Selection (SFFS) for feature selection. Lastly, the selected features fed to the seven classifiers: RF, SVM, DT, Gaussian NB, KNN, LR, and ExtraTrees for classification with LOOCV. The results show that the RF performed excellently achieving an accuracy of 93.10%. Moreover, Kothapalli et al. [118] presented a study for the prediction of ADHD and Dyslexia using an ensemble model. The dataset is obtained from the public database Kaggle, and features relevant to speech impairment and eye movement are tracked from this data using DT. After learning the features are passed to the ensemble model (RF, NB, random subspace) for classification using 10-fold CV and achieved 94% accuracy.

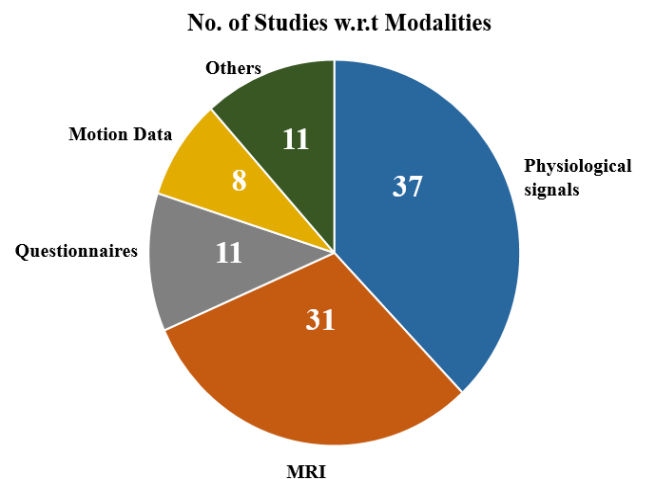
⁴<https://datasets.simula.no/hyperaktiv/>

TABLE 10. Summary of studies that utilize some other diagnostic methods for the diagnosis of ADHD.

Sr.#	Ref.	Modality	Dataset	Age	Features	ML/DL Model	Validation Approach	Accuracy %
1	[114]	Game	66 ADHD, 181 ADHD-Risk, 349 TDCs	8 to 13	time series	RNN series of GRU, RNN, LSTM	LOOCV	97.82
2	[115]	Game	66 ADHD, 181 ADHD-Risk, 349 TDCs	8 to 13	time series	Bidirectional LSTM	LOOCV	98.15
3	[116]	Game	100 ADHD, 100 ASD, 100 ASD + ADHD, 100 TDCs	14 to 18	Behavioral	LSTM, Transformer	60:20:20	71
4	[117]	Pupillometric	28 ADHD, 22 TDCs	10 to 12	Spectral	LR, SVM, KNN	67:33	84.30
5	[118]	Pupillometric	28 ADHD, 22 TDCs	10 to 12	pupil size dilation velocity	LR, KNN, NB, DT, RF, SVM	10-fold CV	85.6
6	[119]	Eye-tracking	56ADHD, 79 TDCs	6 to 12	Eye-tracking	RF, ExtraTrees	5-fold CV	76.3
7	[120]	Handwriting patterns	14 ADHD + ASD, 15 TDCs	5-14, 8-12	statistical	RF, SVM, DT, Gaussian NB, KNN, LR, ExtraTrees	LOOCV	93.10
8	[121]	speech impairment and eye movement.	336066 overall	N/A	speech impairment and eye movement.	Ensamble (RF, NB, random sub space), DT	10-fold CV	94
9	[122]	Power Analysis	1395 overall	N/A	N/A	SVM, NB	80:20	96.75
10	[123]	population registers.	15890 ADHD, 15435 non ADHD	N/A	N/A	LR, DT, RF, XGB,	5-fold CV	79.6
11	[124]	Questionnaire, EEG (multimodality)	306 overall	14-17	ERP	LR, SVM	LOOCV	65

Sai et al. [119] utilized SVM to improve the accuracy of ADHD detection. The methodology involved applying novel SVM on the ADHD dataset consisting of 1395 records and 119 features. The 80:20 split ratio is employed for training and testing purposes. Results show that SVM outperforms for predicting regional ADHD distribution as compared to the Naïve Bayes with 96.75% accuracy. Roche et al. [120] aimed to present an ML stacked model for the diagnosis of ADHD in a non-adult population. The dataset utilized in this study consists of the data from population health care administrative registers encompassing 1225406 non adult individuals, including 15890 ADHD patients and 15435 non-ADHD individuals. Features are extracted from this data and passed to the Chi-square test for feature selection. Finally, the classification is performed using a stacked model (LR, DT, RF, and XGB) on selected features with a 5-fold CV. Results reveal that the proposed model achieves 79.6% accuracy. Furthermore, Hamori et al. [121] evaluate the ERP indices of reinforcement sensitivity as ADHD biomarkers in adolescents. The dataset used in this study is the BLADS study which includes data from baseline and 18-month follow-up assessments of 306 adolescents oversampled for ADHD, along with EEG and questionnaires assessing substance use and ADHD symptoms. ERP features are extracted from this data and the Fisher score is utilized for feature selection. After selecting discriminate features passed to the LR for assessing the variance explained in substance use and SVM for classification with LOOCV. The results indicate the maximum achieved accuracy is 65%.

From TABLE 10, the comprehensive investigation of the studies illustrates that 11 out of 98 studies placed into this

**FIGURE 4.** Graphical representation of ADHD diagnostic methods w.r.t no. of studies.

category. Where 3 studies used game data while 2 employed Pupillometric data. Furthermore, Eye-tracking, handwriting patterns, speech impairment and eye movement, Power Analysis and multimodality (Questionnaire, EEG) are each employed in one research. Additionally, the studies that utilize gaming data as a diagnostic tool employed DL models for the identification of ADHD while all other researches utilize ML models.

The overall summary of all modalities employed for the identification of ADHD diagnosis in selected studies is illustrated in Figure 4.

IV. ANSWER TO THE RESEARCH QUESTIONS

RQ1: Which Diagnostic tools or modalities are primarily utilized for the identification of ADHD and how effective these modalities are?

Among all these diagnostic tools, physiological signals emerged as the most utilized, featured in 37 out of 98 selected studies. These comprise various types of signals such as EEG, ECG and ERG, each recording the electrical activities of the brain. Moreover, MEG signals, which capture magnetic brain activities, and fNIRS which can measure the changes in hemodynamic signals reflecting neural activity are also included. Notable, among all these signals EEGs are the most utilized physiological signals for diagnostic purposes, featured in 30 out of 37 studies. Moreover, physiological signals consistently yield excellent results for the identification of ADHD with the highest achieved accuracy reaching 100%.

Additionally, in almost all studies utilizing these signals, the achieved accuracy exceeds 80%, detail is provided in section III-A. MRI scans are the second most utilized diagnostic tool for ADHD identification with 31 out of 98 studies employing it. MRI scans consist of two types, structural MRI scans help to identify anatomical changes in the brain, and functional MRI scans help to understand how various parts of the brain work together. Significantly, among these types, functional MRI scans utilized alone in 26 out of 31 studies, and 1 study used in conjunction with structural MRI scans as a diagnostic tool. MRI scans also yield outstanding results with a maximum accuracy of 99% and in nearly all studies accuracies exceed 70%, detail is presented in section III-B.

Moreover, questionnaires are the third most utilized diagnostic tool for the detection of ADHD with 11 out of 98 studies employing it. The primary goal of questionnaires is to collect quantitative data. The highest accuracy achieved by questionnaires is also excellent at 99.60% with overall accuracies surpassing 70%, detail is presented in section III-C. Furthermore, motion data is the fourth highest employed diagnostic tool for the diagnosis of ADHD, utilized in 8 out of 98 researches. Among various data recording devices used to collect motion data, the Actigraphy device emerged as the most employed with 5 out of 8 studies utilizing it. In terms of accuracy, the highest achieved accuracy by this diagnostic tool is 99.85%, and nearly in all studies accuracies suppressing 80%, detail is presented in section III-D. Lastly, researchers also employed some other modalities such as gaming data, pupilleometric data, handwriting patterns, etc. However, these modalities are used in either one, two, or three studies. Further, detail is provided in the section III-E. The overall summary is illustrated in Figure 5.

RQ2: What are the primarily employed ML and DL models for the identification of ADHD?

By conducting a systematic review of the literature, we have identified that the utilization of ML approaches is higher as compared to DL approaches. Notably, 45 out of 98 studies employed ML approaches, 33 studies used DL approaches, and 20 researches employed both ML and DL approaches.

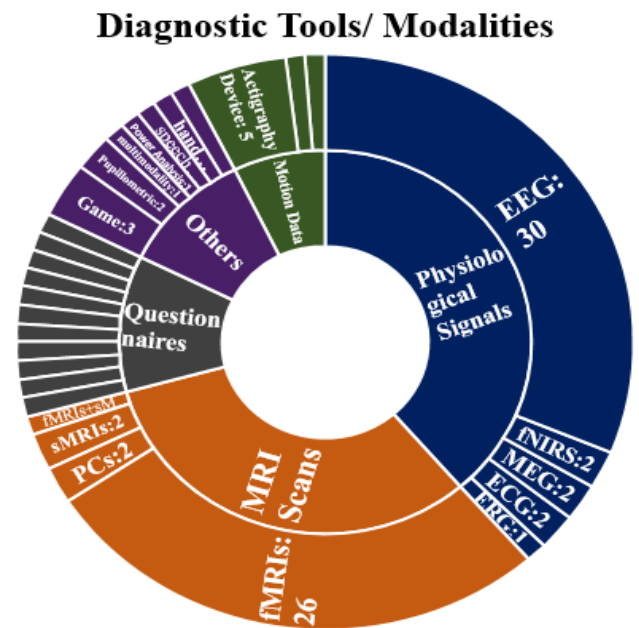


FIGURE 5. Graphical representation of all ADHD diagnostic tools, the first level refers to the diagnostic tools and the second level indicates their subtypes.

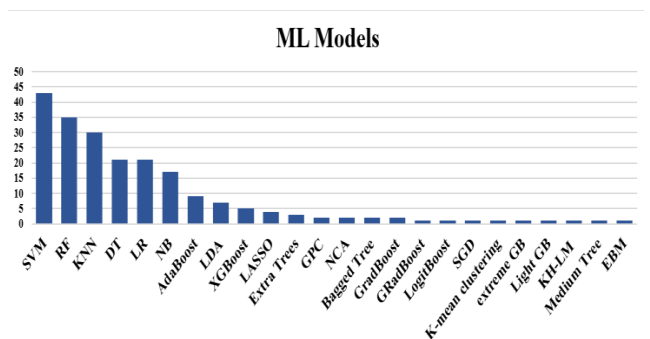


FIGURE 6. Graphical depiction of ML Models utilized for ADHD diagnosis.

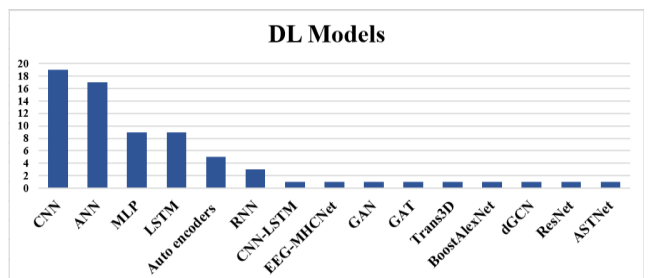


FIGURE 7. Graphical illustration of DL Models employed for ADHD identification.

Among ML models SVM emerged as the most frequently utilized classification model with 43 researches using it for the identification of ADHD. Following SVM, RF ranked as the second most employed, while KNN stands for the third most utilized with 35 and 30 researches respectively. Following the above-mentioned models, DT and LR utilized

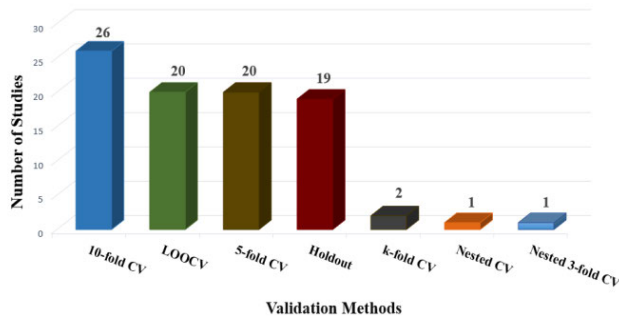


FIGURE 8. Graphical representation of validation methods utilized by AI models for ADHD detection.

in 21 researches each, NB employed in 17 studies, AdaBoost in 9 studies, LDA utilized in 7 researches and XGBoost employed in 5 studies. Moreover, LASSO used in 4 studies, Extra Trees employed in 3 studies, while GPC, NCA, Bagged Tree, GradBoost, and Elastic Net are each utilized in 2 studies. Furthermore, GRadBoost, LogitBoost, Stochastic Gradient Decent (SGD), K-mean clustering, extreme GB, Light GB, KH-LM, Medium Tree and EBM are each employed in 1 study for the diagnosis of ADHD as presented in Figure 6.

On the other hand, in DL approaches CNN is the most frequently employed model, being utilized in 19 research studies. Following CNN, the ANN utilized in 17 studies, and both MLP and LSTM used in 9 studies each. Following the aforementioned models, Auto encoders employed in 5 studies, and RNN in 3 studies. Afterward, CNN-LSTM, EEG-MHCNet, GAN, GAT, Trans3D, BoostAlexNet, ResNet, ASTNet and dGCN are each utilized in 1 study for the identification of ADHD as illustrated in

RQ3: What are the mainly utilized validation approaches to classify ADHD patients from TDCs?

During the thorough analysis conducted for this SLR, we identified various validation approaches utilized for the classification of ADHD patients from TDCs. Among all these approaches 10-fold CV emerged as the most commonly used approach, being employed in 26 research studies. Following the 10-fold CV, LOOCV and 5-fold CV stands for the second most utilized approach, being utilized in 20 studies each. Subsequently, Holdout Method is the third most utilized approach appear in 19 studies. The holdout method utilized various training, validation, and testing splits with the most frequent 80:20 split ratio, appearing in 8 studies. The 70:30 split ratio ranked second most common, being used in 4 studies while all other splits employed in one study each. Following the aforementioned approaches, K-fold CV in which number of folds are not mentioned employed in 2 researches while nested CV and Nested 3-fold CV each utilized in 1 study. The overall results are summarized in Figure 8.

RQ4: Which other comorbidities are considered with ADHD in current research?

Overall, 13 out of 98 selected studies differentiate ADHD from other comorbid conditions with ADHD. The

summarized results of various comorbidities associated with ADHD are presented in TABLE 11 with the following defined parameters. Firstly, **Modality** indicates the diagnostic tool used for neurodevelopmental disorders.

Secondly, **Dataset** refers to the size of data and **Comorbidity** indicates the disorder considered alongside ADHD. Subsequently, the **ML\DL Model** describes the classifiers used for the classification or detection of neurodevelopmental disorders whether ML models, DL models, or both. Moreover, the **Validation Approach** denotes the training, validation, and test split ratios employed with the ML\DL model for classification. Lastly, **Accuracy %** indicates the maximum accuracy achieved by any model in percentage. The detail is presented in subsequent paragraphs.

In TABLE 11, three studies considered to detect Conduct Disorder (CD) as a comorbidity associated with ADHD [15], [30], [54]. Among these Loh et al [30] and Koh et al. [15] employed ECG data, including 45 ADHD patients, 16 CD patients, and 62 patients with both conditions, while Tor et al. [54] utilized EEG data with the same sample size. Additionally, six studies aimed to detect Autism Spectrum Disorder (ASD) as a comorbid condition with ADHD [60], [85], [86], [94], [113], [117]. Among these Manjur et al. [60] employed ERG data, which consists of 94 ASD patients, 46 ADHD patients, and 146 TDCs.

Ji et al. [85] employed three datasets containing fMRI samples along with Personal Characteristics (PC): the first dataset included 319 ADHD patients and 528 TDCs, the second included 527 ASD patients and 549 TDCs, and the third contained 483 ASD patients and 556 TDCs.

Similarly, Hu et al. [86] also utilized three fMRI samples along with PCs: the first dataset contained 358 ADHD patients and 581 TDCs, the second consisted of 505 ASD patients and 530 TDCs, and the third encompasses 521 ASD patients and 592 TDCs. Grazioli et al. [94] used questionnaire-based data including 432 participants who may or may not have ADHD and/or ASD. Jaiswal et al. [113] employed a gamified web system for data collection, encompassing 100 ADHD patients, 100 ASD patients, 100 TDCs, and 100 patients having both conditions. Similarly, Shin et al. [117] used handwriting patterns to get information about ADHD and/or ASD patients, including 14 individuals having both conditions and 15 TDCs.

Moreover, two research studies tried to differentiate Alzheimer's disease from ADHD [83], [84]. Both studies utilized two datasets consisting of fMRI samples but the number of samples differed. Zhang et al. [83] employed a first dataset containing 357 individuals having ADHD and 582 TDCs while a second dataset encompassing 44 individuals having Alzheimer's and 38 TDCs. Wang et al. [84] utilized the first dataset, including 118 individuals with ADHD and 98 TDCs while the second dataset contained 56 people with early Alzheimer's 43 with late Alzheimer's, and 50 TDCs. One study considered several comorbid conditions such as Major Depressive Disorder (MDD), Generalized Anxiety Disorder (GAD), Bipolar Disorder (BD), and Alcohol Use

TABLE 11. Summary of research studies that considered other comorbid conditions with ADHD.

Sr.#	Ref.	Modality	Dataset	Comorbidity	ML/DL Model	Validation Approach	Accuracy (%)
1	[31]	ECG	45 ADHD, 62ADHD+CD, 16 CD	CD	CNN, Grid-CAM	10-fold CV	96.04
2	[56]	ECG	45ADHD, 62ADHD+CD, 16 CD	CD	DT, KNN, AdaBoostM2, Bagged Tree, Subspace, SVM, SFS	10-fold CV	87.19
3	[57]	EEG	45ADHD, 62ADHD+CD, 16 CD	CD	DT, KNN, AdaBoostM2, Bagged Tree, Subspace, SVM, SFS	10-fold CV	97.88
4	[62]	ERG	94 ASD, 46 ADHD, 146 TDCs	ASD	RF, AdaBoost, GradBoost, XGBoost, SVM, KNN, MLP	10-fold CV	70
5	[87]	rs-fMRI + PC	(319 ADHD, 528 TDCs), (527 ASD, 549 TDCs), (483 ASD, 556 TDCs)	ASD	CNN with hash layer	5-fold CV	69.09
6	[88]	rs-fMRI, PC	(358 ADHD, 581 TDCs) (505 ASD, 530 TDCs) (521 ASD, 592 TDCs)	ASD	LSTM, transfer learning	10-fold CV	72.62
7	[97]	Questionnaire	342 overall	ASD	DT	LOOCV	74
8	[116]	Game	100 ADHD, 100 ASD, 100 ASD + ADHD, 100 TDCs	ASD	LSTM, Transformer	60:20:20	71
9	[120]	Hand writing patterns	14 ADHD + ASD, 15 TDCs	ASD	RF, SVM, DT, Gaussian NB, KNN, LR, ExtraTrees	LOOCV	93.10
10	[85]	rs-fMRI	(357 ADHD, 582 TDCs) (44 early MCIs, 38 TDCs)	Alzheimer	Transformer with defision kernel	80:20	78.4
11	[86]	rs-fMRI	(118 ADHD, 98 TDCs) (56 early, 43 late MCI, 50 TDCs)	Alzheimer	Multi kernel SVM	5-fold CV	72.3
12	[98]	Questionnaire	432 ADHD, 523 Non ADHD	MDD, GAD, BD, AUD	Elastic Net	5-fold CV, LOOCV	78.8
13	[121]	speech impairment and eye movement.	336066 overall	Dyslexia	Ensamble (RF, NB, random sub space), DT	10-fold CV	94

Disorder (AUD) associated with ADHD [95]. In this study Liu et al. [95] employed questionnaire-based data, comprising 432 people with ADHD and 523 individuals without this disorder. Furthermore, one study considered Dyslexia as a comorbidity with ADHD [118]. In this study, Kothapalli et al. [118] utilized speech impairment and eye movement features from 336066 participants.

RQ5: What are the gaps in the current research and which key improvements are required to diagnose ADHD more efficiently?

The diagnosis of ADHD is crucial, as this is one of the most frequently diagnosed neurodevelopmental disorders. For this purpose, researchers employed various types of modalities or diagnostic tools. However, there are still some gaps in this area, which are detailed in the following paragraphs regarding the different modalities used.

Physiological signals (TABLE 9) are the most frequently utilized modalities for the identification of ADHD and they yield outstanding results. More specifically EEG signals among all types of signals are the most commonly utilized. ECG, fNIRS, and MEG signals are also important diagnostic tools as they also provide excellent results. However, only two studies have used each of these modalities, indicating a need for further exploration and utilization of these modalities in ADHD research.

According to the above-mentioned information (TABLE 10), MRI scans ranked as the second most employed

modality. The studies utilizing these modalities predominantly employed DL approaches with 21 out of 31 studies employing only DL approaches and 5 combining DL approaches with ML approaches. Despite their effectiveness, the features of deep learning models remain undisclosed to us because of their black-box nature which poses a challenge in the medical domain. In this domain, the possible treatments are driven by informed decisions, not by simple yes/no algorithms. Nevertheless, according to our selection and rejection criteria, none of the 31 studies using this modality employed any method to explain these hidden features, despite the availability and application of several Explainable Artificial Intelligence (XAI) techniques such as Classification Activation Map (CAM) [122], Local Interpretable Model-Agnostic Explanations (LIME) [123] in other medical domains. Among all other modalities, only physiological signals utilized XAI models, even then, only three studies [30], [32], [36] have done so, none of which applied these techniques in detail. Consequently, there is a need to utilize XAI models that focus on the interpretability and explainability of hidden features of DL models, enabling us to better understand the underlying causes of ADHD. These models show which brain regions most influence the model's prediction. As discussed previously, most studies using this modality have employed DL approaches, with CNNs and ANNs being the most common. However, several transfer learning techniques, such as VGG [124] and ResNet [125],

are also available and perform well in image-related tasks. These techniques should also be explored further in the context of ADHD diagnosis.

Motion data or activity recording modalities would be highly desirable because devices used to record this type of data can more efficiently represent symptoms of inattentiveness and hyperactivity in people's daily lives. However, according to our analysis, only 8 studies employed this modality despite its excellent results (TABLE 9). Therefore, there is a need to investigate the usability and efficiency of wearable devices for diagnosing ADHD. Additionally, since most ML models have been utilized with this modality so DL models should also be explored. Moreover, the game skeleton data despite yielding good results has only been employed in two studies (TABLE 10), so there is a need to explore this area further for the diagnosis of ADHD by creating some large datasets. Similarly, handwriting patterns can also be explored further by creating some large datasets. Additionally, current studies often focus on individual modalities; only two studies have employed multi-modal approaches, such as [73], which utilized sMRIs and rs-fMRIs, and [121], which combined questionnaires and EEG. Consequently, research is needed to effectively integrate and analyze data from multiple modalities to improve ADHD diagnosis. Furthermore, despite many studies showing promising results, there remains a lack of real-world implementation of AI-based ADHD diagnostic tools. This highlights the need to develop practical AI-based diagnostic tools for ADHD that can be applied in real-world clinical settings.

V. DISCUSSION AND LIMITATIONS

ADHD is one of the most frequently identified neurodevelopmental conditions. It primarily affects children but also prevails in adults. The prevalence of ADHD has made its identification an active area of research. Therefore, in this study, we conducted an SLR to examine the state-of-the-art research on the diagnosis of ADHD. The primary objective of this SLR is to thoroughly investigate the diagnostic tools or datasets utilized for the identification of ADHD. It also aims to find which AI (ML or\and DL) models are employed for the detection or classification of individuals with ADHD from TDCs, and to determine the key improvements needed in this area. To achieve these objectives, we select ninety-eight (98) recent studies out of 2452 studies based on inclusion and exclusion criteria. Subsequently, we grouped the selected studies into five categories based on the diagnostic tools or modalities these studies employed i.e. physiological signals, MRI, questionnaires, motion data, and others category. These categories have been evaluated based on various parameters. A detailed investigation of these modalities along with AI models provides ease to the researchers in the selection of appropriate modalities or data and AI models for the diagnosis of ADHD.

The categorization of diagnostic tools used for the diagnoses of ADHD is illustrated in TABLE 8. Among

the ninety-eight (98) selected studies, thirty-seven (37) researches placed into the physiological signals category. These studies utilized various types of physiological signals as a modality or dataset for the identification of ADHD, i.e. EEG, ECG, ERG, MEG, and fNIRS signals. Specifically, the majority of these studies focus on EEG signals with thirty (30) studies utilizing this modality. Thirty-one (31) studies are classified under the MRI category. These studies employed fMRI scans performed at resting state and sMRI scans for the diagnosis of ADHD. Notably, rs-fMRI scans have been used in most of the studies, being employed in twenty-six (26) studies. Additionally, eleven (11) research studies fall into the questionnaires category, as they utilized various types of questionnaires or rating scales to gather data from ADHD patients and TDCs. No specific questionnaire is deemed more important as compared to others; all are considered equally significant. Details are provided in TABLE 9. Moreover, eight (8) studies are classified into motion data or activity recording category. These studies collect data by observing individual's daily routines with the help of devices. Specifically, the Actigraphy device is used to gather data, with five (5) studies employing this device. Lastly, eleven (11) studies are placed into the others category. The studies use less common modalities for data collection purposes. These modalities are either employed in one, two, or a maximum of three studies details are given in TABLE 10. Furthermore, the investigation of AI models indicates that ML models are utilized more frequently, with forty-five (45) researches employing only ML models and twenty (20) studies combining ML models with DL models. Notably, among ML models, SVM is the most frequently employed appearing in forty-three (43) studies followed by RF and KNN being employed in thirty-five (35) and thirty (30) studies. Subsequently, only DL models are used in thirty-three studies (33) studies, with the majority utilizing CNN in nineteen (19) researches followed by ANNs being employed in seventeen (17) studies. Besides AI models we also investigate various validation methods utilized by these models. Among them, 10-fold CV emerged as the most frequently employed validation method with twenty-six (26) studies using it followed by LOOCV and 5-fold CV being employed in twenty (20) studies each. As ADHD is a heterogeneous disorder and it can also co-occur with other neurodevelopmental disorders. In this SLR, we also analyze which other comorbid conditions have been considered alongside ADHD. Thirteen (13) studies have been identified that explore other comorbidities, with ASD being the most frequently associated condition with ADHD. Notably, six (6) studies have examined this comorbid condition with ADHD.

The comprehensive analysis conducted in this SLR reveals several limitations that exist in this area of research. Despite physiological signals being the most commonly employed modalities, the availability of publicly accessible datasets is limited except for EEG. While the dataset of MEG signals is available but remains underexplored have been utilized in only two studies. Moreover, no publicly available datasets

exist for other types of signals. Therefore, there is a need to create large size of physiological signal datasets and evaluate various ML and DL models using these datasets. MRI scans are the second most frequently used modality, mainly using DL models for classification or detection. DL models are often considered as black-boxes, making it difficult to understand their internal workings. When dealing with clinical data it is crucial to know what is happening inside to make informed decisions. To the best of our knowledge, none of the studies using MRIs have utilized explainable DL models. Furthermore, across all other modalities, XAI models have only been applied to physiological signals which are also limited to three studies [30], [32], [36], and none of them applied in detail. To address this issue researchers, need to explore and utilize more explainable and interpretable AI models to uncover the hidden features of DL models, helping us to understand the underlying causes of ADHD. As discussed before most studies using this modality rely on DL approaches like CNNs and ANNs, but transfer learning techniques like VGG [124] and ResNet [125], which excel in image-related tasks, should also be further explored. Moreover, activity recordings can provide more effective information regarding ADHD symptoms by analyzing an individual's daily routines. However, only eight (8) studies employed these modalities with few devices used for recording purposes, particularly the Actigraphy device. Therefore, there is a need to investigate other activity-recording devices that can collect motion data more effectively. Some less common modalities such as game skeleton data and handwriting patterns also shown promising results and should be further explored. Additionally, most of the studies have relied on individual modalities for ADHD diagnosis, except two studies [73], [121]. This highlights the need to explore multi-model data to improve ADHD diagnosis. Moreover, despite many studies demonstrating excellent results in diagnosing ADHD, there is still lack of real-world implementation of AI-based tools. Consequently, it is essential to develop automated AI-based ADHD diagnostic tools that can be applied in real-world settings.

This study still has some shortcomings.

- Five popular scientific databases are selected such as Google Scholar, IEEE, Springer, Science Direct, and PubMed. These databases provide a huge amount of state-of-the-art research studies, particularly journal papers. Consequently, our search process is based on these databases. However, some related studies may exist in other scientific databases. Therefore, there is a chance we may have missed some relevant studies. Nevertheless, we hope that the rejection of such databases does not notably affect our results.
- We have chosen to include only English-language research studies. However, this may result in the omission of relevant research written in other languages, such as French or German.

Comparing the results of ADHD diagnosis studies is challenging because they employ different modalities. Even when

the same modality is utilized the number of subjects varies. Moreover, these research studies normally have different data acquisition and preprocessing methods. So, we cannot make a comparison. Specifically, an SLR is conducted to find and analyze 93 research studies published between 2021 to 2024.

VI. CONCLUSION

This article thoroughly examines various ADHD diagnostic tools and broadly includes the research studies that employed AI approaches for ADHD detection or classification. These studies further partitioned into five categories w.r.t the modality or dataset they employed: Physiological Signals, MRI, Questionnaires, Motion Data, and Others. Subsequently, a critical analysis is carried out to investigate how these modalities help in the detection of ADHD. The analysis reveals that primarily EEG signals and fMRIs are employed for the identification of ADHD and few studies utilized other modalities, there is a need to explore other modalities. Additionally, there is a lack of publicly available datasets for modalities except for MRI and EEG. Datasets for MEG and actigraphy do exist but remain underexplored. Consequently, we suggest that more publicly available datasets need to be created for modalities such as ECG, Game data, and data collected through various recording devices. With the growing applicability of DL models, transfer learning approaches should also be further explored. Moreover, in the medical domain, the interpretability and explainability of AI models, particularly DL models, are essential for making informed decisions. Therefore, we recommend the development of interpretable AI models, specifically DL models, for the diagnosis of ADHD. Most studies focus on individual modalities for ADHD diagnosis, underscoring the need to explore multi-model data. In spite of promising results, AI-based diagnostic tools are still not widely implemented in real-world settings, underscoring the need for automated tools that can be practically applied.

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LIST OF ABBREVIATIONS

SLR:	Systematic Literature Review.
ADHD:	Attention Deficit Hyperactivity Disorder.
TDCs:	Typical Developmental Controls.
AI:	Artificial Intelligence.
ML:	Machine Learning.
DL:	Deep Learning.
MRI:	Magnetic Resonance Imaging.
EEG:	Electroencephalogram.

ECG:	Electrocardiogram.
ERG:	Electroretinogram.
MEG:	Magnetoencephalography.
FNIRS:	Functional near-infrared spectroscopy.
SVM:	Support Vector Machine.
PCA:	Personal Component Analysis.
LR:	Logistic Regression.
RF:	Random Forest.
KNN:	K Nearest Neighbor.
EGB:	Extreme Gradient Boosting.
Grad-CAM:	Gradient-weighted Class Activation Mapping.
NB:	Naïve Bayes.
CNN:	Convolutional Neural Network.
MLP:	Multi-Layer Perceptron.
DT:	Decision Tree.
LASSO:	Least Absolute Shrinkage and Selection Operator.
EBM:	Explainable Boosted Machine.
LRP:	Layer-wise Relevance Propagation.
LDA:	Linear Discernment Analysis.
ANN:	Artificial Neural Network.
NCA:	Neighborhood Components Analysis.
LSTM:	Long Short-Term Memory.
GLM:	Generalized Linear Model.
CFS:	Correlation-based Feature Selection.
HMMs:	Hidden Markov Models.
SFS:	Sequential Feature Selection.
ResNet:	Residual Network.
mRMR:	Minimum Redundancy Maximum Relevance.
DGAN:	Deep Generative Adversarial Network.
Att-AENet:	Attention-Aware Encoder-Decoder Network.
SSANN:	Summation-based Synergic Artificial Neural Network.
ASTNet:	Attention-based Spatio-Temporal Network.
RNN:	Recurrent Neural Network.
RLR:	Regularized Logistic Regression.
KSADS:	Kiddie Schedule for Affective Disorders and Schizophrenia.
DIVA:	Diagnostic Interview for ADHD.
CPRS-R:	Conner's Parent Rating Scale-Revised.
CBCL:	Child Behavior Checklist.
SRS:	Social Responsiveness Scale.
CTRS-R:	Conner's Teacher Rating Scale-Revised.
VADRS:	Vanderbilt ADHD Diagnostic Rating Scale.

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